Statement E: An Explicit Exterior Native Map

A finite event construction and a direct canonical-class proof

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Abstract

We prove Statement E of the supplied problem brief, using its expressly granted analytic word, native-complex, and binary-seam inputs. A finite projection of a suitably chosen nearby normalization gives a transitive factorization of the central event word into conjugated local branch words and positive single generators. Explicit signed word-grid relations, a finite Hurwitz algorithm, and star scissors turn this into an exact seam of marked factor grids. We give the protected-summand cancellation rules needed to remove every conjugator.

The local maps are constructed directly on the native complexes over $\mathbb{F}_2[T, T^{-1}]$. A rectangle identity gives a polynomial degree-zero detecting equivalence for each degree-minus-one O switch. Component derivatives turn the global class calculation into the determinant of an explicitly chosen rooted-tree incidence matrix. Explicit auxiliary contractions and an explicit simple-factor functional finish the construction. The resulting map has Maslov degree zero, shifts $(0, \mu(V)/2)$, and sends the canonical class to the canonical class. All additional analytic and combinatorial lemmas are proved, including the native maps used on intermediate auxiliary marking pairs. The document includes exact seam formulas, the full operation count, exceptional cases, a fully indexed $S = 4, \mu(V) = 1$ example, and a dependency ledger.

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1 Statement E and the inputs used

We use the notation and the explicitly granted inputs of the supplied problem brief [1]. An *assembly* and a *level* mean the data specified there, including their invariant identities. The assertion proved here is the chain-map statement itself.

Theorem 1 (Statement E)**.** *Let $\boldsymbol{n} : (\mathbb{C}^2, 0) \to (\mathbb{C}^3, 0)$ be a holomorphic map germ with an injective representative and $(d\boldsymbol{n})_0 = 0$. There exist an assembly, an allowed nonzero level s_0, a positive word $\mathbf{w}$ on S strands, positive words β_j on a'_j strands for the enumerated singular points Q_j, and grids D_j such that*

$$\begin{aligned} &\operatorname{cycles}(\pi(\mathbf{w})) = b, && \ell(\mathbf{w}) + S(S-1) + b = 2\delta(W) + S, \\ &2 \le a'_j \le t_J, \qquad \operatorname{cycles}(\pi(\beta_j)) = 1, && \ell(\beta_j) + 1 = 2\delta_j + a'_j, \end{aligned}$$

each β_j contains every generator, and $\operatorname{wordGrid}(\beta_j) \sim D_j$. Every D_j has size at least two. They are seam prepared, and there is an $\mathbb{F}_2$-linear chain map in each Maslov degree

$$\psi : C\big(\operatorname{wordGrid}(\mathbf{d}_S\mathbf{w})\big) \longrightarrow C\big(\operatorname{iteratedSeamGrid}(D)\big)$$

of Maslov degree zero, with bifiltration shifts $(0, \mu(V)/2)$, for which

$$\psi_* \operatorname{can}\big(\operatorname{wordGrid}(\mathbf{d}_S\mathbf{w})\big) = \operatorname{can}\big(\operatorname{iteratedSeamGrid}(D)\big).$$

One can take $a'_j = a_j$. More strongly, the constructed D_j satisfy $X(0) = 0$, $O(\text{last}) = 0$, and $X(\text{last}) = \text{last}$ simultaneously. Thus all row rotations in the final iterated seam are the identity.

The proof constructs the map directly. Some intermediate homogeneous maps have degrees -1 and $+1$; their composite has degree zero. Auxiliary pairs of marking permutations are used for the formulas of those maps. The source and target in the theorem are valid native grids. This is an alternative to the restricted M/X/B movie in Section 3.4 of the brief. In particular, it makes no claim to be a movie consisting only of those three permitted kinds.

Here is the numerical structure of the construction. A finite event factorization has N singular factors and B positive single-generator factors. Put

$$r = N + B, \qquad s = (S-1)(r-1), \qquad U = \sum_{j=1}^{N} (S - a_j) + B(S-2). \tag{1}$$

The analytic calculation proves

$$B = S - \sum_j a_j + N - 1 + \mu(V), \qquad s - U = \mu(V). \tag{2}$$

The scissor construction has $s - U$ X switches and U O switches. The O switches are paired algebraically with U auxiliary contractions. A determinant calculation, proved below from rectangle counts, gives the canonical-class identity for this complete composition. A separate degree-zero normalization of each four-mark auxiliary block contributes no filtration shift. Finally an explicit functional removes each single-generator tensor factor.

1.1 The admitted inputs

We use the following statements precisely in the form granted in Sections 2 and 5 of the brief:

1. The assembly and arbitrarily small allowed levels exist; $v = z_0 \circ \boldsymbol{n}$ is reduced; the nearby normalization is smooth; its singular image germs are unibranch with $2 \le a_j \le t_J$; $N \ge 1$; and

$$2\delta(V) + M = S, \quad \delta(W) = \delta(V) + \sum_j \delta_j, \quad \mu(V) = 2\delta(V) - b + 1, \quad S \ge 4.$$

   The branch count of the source and central image is the same b.

2. The central and local *event-word* constructions (I1), (I2), (B1), and (B2) are available for transverse projections. Thus the central word is $\mathbf{d}_S\mathbf{w}$, with $\mathbf{w}$ positive of cycle count b, and

$$\ell(\mathbf{d}_S\mathbf{w}) = 2\delta(W) + S - b, \qquad \ell(\beta_j) = 2\delta_j + a_j - 1.$$

   The local β_j is a positive word with one cycle and every generator. We use these event words, rather than an inference from ambient link isotopy.

3. The native rectangle differential squares to zero. The degree-zero polynomial homology image of every valid grid is one dimensional and nonzero. The given native commutation and cyclic permutation maps have degree zero, shifts $(0, 0)$, and preserve its nonzero element. The supplied combinatorial corner reduction expresses the other stabilization orientations using the primitive orientations, commutations, and cyclic permutations.

4. The binary seam–tensor comparison is a filtered chain homotopy equivalence of degree zero and shifts $(0, 0)$ in both directions, with the stipulated canonical-class identity. No naturality of that comparison for derivatives of O variables is assumed or used.

All projection, event-factorization, signed-word, marked-cancellation, seam-coordinate, auxiliary-map, and class-pairing assertions beyond these inputs are proved below. Weierstrass preparation and division, continuity of polynomial roots, the implicit function theorem, Puiseux parametrization, and the one-variable maximum principle are the elementary analytic tools permitted by the brief. The explicit compact vector-field flows used below also give their own local trivializations.

2 Conventions and native maps along ordinary moves

All coefficients are in $\mathbb{F}_2$ and $\Lambda = \mathbb{F}_2[T, T^{-1}]$. A size-$n$ grid is specified by permutations O, X of $\{0, \dots, n-1\}$, with $O(c) \neq X(c)$ and $n \geq 2$. A marking in square (c, r) has centre $(c + \frac{1}{2}, r + \frac{1}{2})$. A state x is a permutation, represented by the lattice points $(c, x(c))$. For finite sets,

$$I(P, Q) = \#\{(p, q) : p_1 < q_1,\ p_2 < q_2\}, \qquad J(P, Q) = \frac{I(P, Q) + I(Q, P)}{2}.$$

Write

$$M_O(x) = J(x, x) - 2J(x, O) + J(O, O) + 1, \qquad A(x) = \frac{M_O(x) - M_X(x)}{2} - \frac{n-1}{2}.$$

The complex is the free Λ-module on the states, with

$$dx = \sum_y \sum_{q \in \mathrm{Rect}^\circ(x,y)} T^{n_O(q)} y, \qquad \deg(T^k x) = M_O(x) - 2k.$$

Here the southwest and northeast corners of a rectangle from x to y belong to x, its other corners belong to y, and its interior contains no point of x. Rectangles containing X markings are counted. The two filtration coordinates of $T^k x$ are $(-k, A(x) - k)$. The polynomial subcomplex P is spanned by the monomials with $k \geq 0$. Its nonzero degree-zero homology image in C is denoted can.

The following two rectangle formulas follow by the four-corner indicator calculation in J:

$$M_O(x) - M_O(y) = 1 - 2n_O(q), \qquad A(x) - A(y) = n_X(q) - n_O(q) \tag{3}$$

for an empty rectangle $q : x \to y$. For a nonempty rectangle, the first right-hand side has the additional term $2\#(\mathrm{Int}\, q \cap x)$. To verify the formulas, replace the two diagonal state points by the other diagonal. A state point inside contributes two to the $J(x, x)$ difference; a marking inside contributes two to the $2J(x, O)$ difference; points outside contribute zero. The moving corners contribute one. Choose cuts outside a torus rectangle to obtain the same calculation for wrapping

rectangles. Transposing both coordinates preserves these formulas and the gradings, with the state permutation replaced by its inverse.

An *auxiliary pair* permits $O(c) = X(c)$. Its complex and gradings use the same formulas. Its canonical class is well defined whenever $n \geq 2$: choose the disjoint permutation $Y(c) = O(c+1)$ and identify its O complex with the valid grid (O, Y). The complex, degree, polynomial subcomplex, and resulting class do not depend on this choice. Auxiliary pairs are used only for explicitly given algebraic maps; the theorem's endpoints are valid grids.

Lemma 2 (Polynomial nonvanishing; uses the canonical-class input)**.**

Proposed name: `polynomialImageNonzero`.

A degree-zero chain map with first shift zero sends the canonical class either to zero or to the target's canonical class. Thus nonvanishing of that image suffices for the required class identity.

Proof. The map takes P into P and therefore takes the degree-zero polynomial homology image into that of the target. The latter has exactly one nonzero element. The same argument applies to auxiliary pairs by their O-complex identification. □

Lemma 3 (Transport along ordinary moves; uses the native move inputs)**.**

Proposed name: `movesToNativeMap`.

Every explicit finite ordinary-move sequence between grids of size at least two gives Λ*-linear maps in both directions of degree zero, shifts* $(0, 0)$*, and canonical-class identity. After O markings are coloured constantly on each component, the maps may be chosen to intertwine, on homology, the sums of the O derivatives of every colour. A stabilization gives its new O the colour of its component.*

Proof. For the degree, filtration, and class assertions, apply the granted maps to each commutation and cyclic permutation. For a primitive stabilization use the explicit maps verified in Section 6; for another corner apply the supplied combinatorial corner reduction and compose those maps. Reverse the reduction to obtain the other direction. Induction on the length of the sequence proves the assertions. The separate-variable proof in Section 6 verifies the derivative claim for the primitive maps, commutations, and permutations. Its differentiated identities compose, so the same induction proves that assertion. This proves the map version of the corner reduction and the finite-sequence assertion M702; no invariance theorem for isotopic links is used. □

For completeness, the seam of sizes $n+1, m+1$, under the conditions $X_G(n) = n$, $X_H(0) = 0$, $O_H(m) = 0$, is the size-$n+m+1$ pair

$$\Sigma.Y(c) = \begin{cases} G.Y(c), & c < n, \\ n + H.Y(d), & c = n + d,\ H.Y(d) \neq 0, \\ G.Y(n), & c = n + d,\ H.Y(d) = 0, \end{cases} \qquad Y \in \{O, X\}. \tag{4}$$

For arbitrary prepared factors, at an intermediate size q the specified row rotation is addition of $q - 1 - X(\text{last})$ modulo q. The iterated size is $\sum_j \text{size}(D_j) - (N-1)$. Our stronger preparation makes each of these additions zero.

Generators have indices starting at zero and $\rho_m = \sigma_0 \cdots \sigma_{m-2}$, $\mathbf{d}_m = \rho_m^m$. The positive letter σ_r has the three-column marking heights

$$O = (r + \tfrac{1}{2}, r, r+1), \qquad X = (r, r+1, r+\tfrac{1}{2}).$$

The negative letter uses $O = (r + \frac{1}{2}, r+1, r)$, $X = (r+1, r, r+\frac{1}{2})$. Integer-height arcs join successive occurrences of that rank cyclically; the middle arc joins the first and third column of its letter. Rows are ordered by their heights, with a specified tie order. This recovers the positive

word grid of the brief up to its allowed tie changes. A completely unused rank, in the signed extension, is retained as an explicit four-mark 2×2 component when it arises by cancellation. All initial word normalizations will retain every generator.

Tracing a component through either three-column gadget exchanges its two integer ranks. Consequently the components of a closed signed word grid correspond to the cycles of its word permutation. This can also be checked by iterating $c \mapsto X^{-1}(O(c))$: the internal middle arc is traversed once and the next integer arc is precisely the outgoing arc of the transposed rank. This elementary incidence observation will be used to count the auxiliary components.

3 A finite projection and its explicit event factorization

Lemma 4 (Finite event factorization; new analytic)**.**

Proposed name: `finiteProjectionEventFactorization`.

For an assembly as in Theorem 1, there are an allowed level $s \neq 0$ and a transverse projection h such that a connected compact normalization level has a proper finite projection of degree S, unramified at its boundary and containing all the level's singular points. There is a finite signed-word derivation, up to whole-word conjugacy, of

$$\mathbf{d}_S\mathbf{w} \;\sim\; \prod_{l=1}^{N+B} \gamma_l \eta_l \gamma_l^{-1},$$

where the cores consist of the N supplied local positive event words and B positive single generators. The permutations of the conjugated factors $\gamma_l \eta_l \gamma_l^{-1}$ generate a transitive group, and $B = S - \sum_j a_j + N - 1 + \mu(V)$. All projection, level, path, and finite-event choices are made by the rules in the proof.

Proof. We prove the analytic assertions needed for the event construction. In particular we do not require that a single linear projection have only simple critical points or distinct critical values. Ordinary higher ramification will be replaced by a finite local family of simple ramifications, at the level of boundary configurations.

3.1 An elementary local finite-map fact

We first spell out the finite-map consequence of Weierstrass preparation that will be used. Suppose $F = (f, g)$ is holomorphic near a point of $\mathbb{C}^2$ and that its fibre through that point is isolated there. Then F is open at that point and has finite fibres on a sufficiently small representative. Translate the point and its value to zero, and choose source coordinates (u, w) in which f is regular in w. Preparation, with s as parameter, writes $f(u, w) - s$ as a unit times a monic polynomial $P(u, s; w)$ of some degree p. Divide $g(u, w) - t$ by P, obtaining a polynomial remainder $Q(u, s, t; w)$ of degree less than p. Multiplication by Q in the free module with basis $1, w, \ldots, w^{p-1}$ modulo P is a $p \times p$ matrix. Its determinant $R(u, s, t)$ vanishes exactly when P and Q have a common root. This last assertion follows by factoring P over $\mathbb{C}$: multiplication by Q is invertible in each factor $\mathbb{C}[w]/(w - a)^e$ exactly when $Q(a) \neq 0$.

The isolated-fibre assumption says that $R(u, 0, 0)$ has an isolated zero at $u = 0$. A second preparation therefore writes R as a unit times a monic polynomial in u, with coefficients holomorphic in (s, t). For every sufficiently small (s, t), its roots remain near zero by continuity of roots. Each is accompanied by a common root of P, Q, also near zero. Thus every sufficiently small target point has a preimage in any prescribed sufficiently small source neighbourhood. The same two monic equations bound the number of points in a fibre. This proves the asserted local openness and finiteness, using only preparation, division, and continuity of polynomial roots.

We will also use the elementary observation that a compact analytic subset of an open subset of $\mathbb{C}^2$ is finite. If it had a positive-dimensional irreducible component, choose a coordinate linear function nonconstant on that component and a point where its absolute value is maximal. A local curve branch through that point, parametrized by preparation or Puiseux expansion, contradicts the one-variable maximum principle. A function constant on a local branch of an irreducible component is constant on that component by analytic continuation, so the chosen nonconstant function supplies such a nonconstant branch. A two-dimensional component contains an open set, where the usual maximum principle gives the same contradiction. Finally, a compact zero-dimensional analytic set is finite, since its points are isolated and a finite collection of their isolating neighbourhoods covers it.

3.2 A proper representative with the correct number of sheets

Put $v = z_0 \circ \boldsymbol{n}$. Choose a target-section coordinate ξ_0 transverse to all the central branches, and a complementary coordinate ζ. Write $h_0 = \xi_0 \circ \boldsymbol{n}$ and $k = \zeta \circ \boldsymbol{n}$. The common zero of (v, h_0) is isolated: on $v = 0$ its image lies in $W \cap \{\xi_0 = 0\}$, whose germ is just the origin by transversality; injectivity then gives the unique preimage.

Choose a closed Euclidean source ball $\overline{B}_R$ inside a representative on which $\boldsymbol{n}$ is injective, with zero as the only common zero of (v, h_0). Choose it also inside a representative of the finitely many curve branches and isolated points of the critical-set germ of $\boldsymbol{n}$; the finite-branch assertion is proved below. We also take it inside the isolated-critical-point representative of v, justified below. Thus those later arguments will not require changing this already fixed ball. Then

$$\eta = \min_{\partial B_R} \max(|v|, |h_0|) > 0.$$

There is no additional boundary-choice assertion hidden here: the boundary is compact and contains no common zero. Choose positive target radii smaller than η. The set

$$\Omega = \{p \in B_R : |v(p)| < \epsilon_0, |h_0(p)| < \rho_0\}$$

maps properly to $\Delta_{\epsilon_0} \times \Delta_{\rho_0}$, since the preimage of any compact subset is a closed subset of $\overline{B}_R$ disjoint from its boundary. Its fibres are compact analytic subsets and hence finite. Local openness follows from the preceding Weierstrass argument.

Each connected component of Ω is open and closed in Ω. Its image is open by local openness and closed by properness. Thus every nonempty component maps onto the connected target bidisc and contains a point above $(0, 0)$. Since that fibre consists only of zero, Ω is connected and the map is onto.

For a small nonzero ξ_0, the central curve has exactly S distinct points above it. Indeed its transverse Weierstrass degree is its multiplicity S, and on each normalized branch $\xi_0 = t^m u(t)$ with $u(0) \neq 0$; after replacing t by $t\,u(t)^{1/m}$, this is t^m. Its derivative is nonzero off zero, and the distinct central branches do not intersect away from zero after shrinking. Injectivity identifies these S points with their S source preimages. At each such source point, $dv \neq 0$. The restriction of $\boldsymbol{n}$ from the smooth source branch to the smooth image branch is locally an injective holomorphic map of one complex variable. Its first nonzero power must have exponent one, since a higher power has more than one nearby preimage by continuity of roots. Its derivative is therefore nonzero. The transverse Puiseux calculation now gives $dh_0|_{\ker dv} \neq 0$, so $d(v, h_0)$ is invertible.

Here dv has no zero in a sufficiently small punctured source neighbourhood. To verify this from reducedness, a positive-dimensional branch of the analytic set $\{dv = 0\}$ would have v constant, hence zero, on its normalization. It would therefore be contained in the singular locus of the reduced plane curve $v = 0$. That singular locus is isolated: otherwise a factor of the reduced equation would divide both its partial derivatives, contrary to squarefreeness in Weierstrass

preparation. This justifies choosing the initial ball inside a representative with no other isolated critical points.

Fix $0 < \rho < \rho_0$ and a small closed annulus around $|h_0| = \rho$. The inverse image of that annulus at $v = 0$ is compact and $d(v, h_0)$ is invertible there. Properness and a convergent-subsequence argument therefore allow ϵ_0 to be decreased so that this invertibility holds on the annulus for all $|v| \leq \epsilon_0$. In particular all the vertical-boundary fibres are unramified, and their cardinality is S, by the implicit function theorem and properness, starting with their known cardinality at $v = 0$.

At this point, before choosing the nonzero level, choose a smooth central point p with $0 < |h_0(p)| < \rho/4$. The implicit function theorem for v gives a section $\sigma(s)$ through p. Fix a smaller radius $\epsilon > 0$ on which it is defined, with $|h_0(\sigma(s))| < \rho/2$ on the closed disc of that radius. Decrease the preceding representative radius to this ϵ now. This order of choices ensures that a later choice of an allowed nonzero level need not be changed in order to obtain a section.

3.3 Capturing the singular points and choosing the final direction

The critical set of $\boldsymbol{n}$ is a proper analytic subset of the source: if all its 2×2 minors vanished identically, the local rank theorem on the open rank-one locus would produce positive-dimensional fibres, contradicting injectivity. Its germ has only finitely many curve branches. The function v is nonconstant on each such branch C. Indeed, on its normalization the map $\boldsymbol{n}|_C$ is nonconstant and generically has nonzero derivative. At those points the rank of $d\boldsymbol{n}$ is one and its image is spanned by the tangent to $\boldsymbol{n}(C)$. If $v|_C$ vanished identically, z_0 would annihilate that image, giving $dv = 0$ along C, contrary to the isolated critical point of v.

On a normalized critical branch, therefore, $v = t^q a(t)$ with $q \geq 1$ and $a(0) \neq 0$. For every fixed small parameter-disc annulus its absolute value has a positive minimum. Consequently every intersection of these branches with $v = s$ tends to zero as $s \to 0$. By the choice of the source ball above, these are all the critical points of $\boldsymbol{n}$ in that representative. In particular, by choosing an allowed nonzero level sufficiently small, all of the singular points listed in that level lie in B_R and satisfy $|h_0| < \rho/4$. None is lost when the finite representative is used. There are no additional singular points, since the restriction is injective and is the normalization, as assumed in the question.

Fix such a level s. If necessary replace h_0 by $h = h_0 + \lambda k$, with λ arbitrarily small. There are only finitely many forbidden values of λ: those for which the new projection vanishes on a central tangent or on the tangent of one of the finitely many branches at the Q_j. Every nonzero tangent vector excludes at most one λ, since (ξ_0, ζ) are coordinates. Choose λ outside that finite set.

All the representative properties can be retained with the already chosen level s. Explicitly, k and $\det d(v, k)$ are bounded on $\overline{B}_R$, while $\det d(v, h_0)$ has a positive lower bound on the compact vertical-boundary annulus. Choose $|\lambda|$ so small that $h - h_0$ is smaller than its annular collar width and $\lambda \det d(v, k)$ is smaller than half that lower bound. Then $d(v, h)$ is still invertible on $|h| = \rho$ for the same small range of v. Also require $|h - h_0| < \eta - \rho$ on $\overline{B}_R$. At a boundary point with $|v| \leq \epsilon < \eta$ we have $|h_0| \geq \eta$, and hence $|h| > \rho$. Thus no preimage of the closed target bidisc meets ∂B_R. Properness follows exactly as before. Transversality on the central branches gives a unique common zero of (v, h): their Puiseux units remain nonzero uniformly for sufficiently small λ. Compact fibres are finite, so the same openness and connectedness arguments apply to the new Ω. The central sheet count is still S. This small change does not remove any Q_j from the representative. Also require $|\lambda|$ small enough that the already chosen section still satisfies $|h(\sigma(s))| < 3\rho/4$ on the closed base disc. This is possible by compactness of its image. We henceforth use h and this representative, with the previously fixed ϵ and the previously chosen nonzero level.

At every Q_j, transversality gives local degree exactly a_j. We impose no condition that ordinary critical points be simple or that their values be distinct.

3.4 Connectedness of the compact level

Distinguish the open set Ω from the set with its vertical boundary adjoined:

$$E = \{p \in B_R : |v(p)| < \epsilon, |h(p)| \le \rho\}, \qquad E_s = v^{-1}(s) \cap E.$$

The fibres E_s are compact. For $s \neq 0$ they are smooth complex curves with smooth boundary, because $dv \neq 0$ and $d(v,h)$ is invertible near $|h| = \rho$. The assertion of compactness here refers to E_s, not to the open fibre in Ω.

We give the needed local trivializations explicitly. In source coordinates z_1, z_2, put

$$Y_0 = \frac{(\overline{v_{z_1}}, \overline{v_{z_2}})}{|v_{z_1}|^2 + |v_{z_2}|^2}, \qquad Y_b = \frac{(h_{z_2}, -h_{z_1})}{v_{z_1} h_{z_2} - v_{z_2} h_{z_1}}.$$

The first is defined away from the zero level, and the second is defined on the vertical-boundary collar. Both satisfy $dv(Y) = 1$, and $dh(Y_b) = 0$. Choose a smooth function of $|h|$ equal to one near ρ and zero outside that collar, and use it to interpolate between Y_0 and Y_b. The resulting vector field Y satisfies $dv(Y) = 1$ and is tangent to the vertical boundary. For a short smooth base path γ, integrate

$$\dot{p}(t) = \gamma'(t) Y(p(t)).$$

On a compact strip over a closed subdisc avoiding zero this vector field and its derivatives are bounded. Its solutions therefore exist for the required sufficiently short interval; they cannot leave through the vertical boundary, and reversing the path gives the inverse. These flows identify nearby compact fibres and their boundaries. A finite subdivision handles any compact base path in the punctured disc. In particular the finite sets of their connected components are locally constant and have unique transport along paths.

The open set $\Omega \setminus \{v = 0\}$ is path connected. Indeed Ω is connected. Connect two points by a chain of convex balls inside it, choosing intermediate points outside $v = 0$. On a complex line through two such points the restriction of v is not identically zero. It has only finitely many zeros near the compact segment, and small arcs in that complex line detour around them. Performing this operation in the finitely many balls gives a path in the complement.

The section $\sigma(s)$ was chosen before the nonzero level and before the small projection perturbation. By the explicit order of choices above it is defined on the *entire* disc Δ_ϵ and stays in Ω for the final projection. No further shrinking, and no change of the selected nonzero level, is required here.

Now fix a nonzero base value s_*. Any component of E_{s_*} contains an interior point. Join that point to $\sigma(s_*)$ by a path in $\Omega \setminus \{v = 0\}$. Its projection is a loop in Δ_ϵ^*. Let C_x be the component of the chosen initial point, let C_0 contain $\sigma(s_*)$, and let T be the bijection of component sets supplied by transport around this loop. The chosen path proves $T(C_x) = C_0$. The path obtained by applying the section σ to the same base loop proves $T(C_0) = C_0$. Injectivity of T gives $C_x = C_0$; no equality of their initial components has been assumed. This proves that E_{s_*} is connected. The argument uses a section on the entire punctured disc, so no classification of its fundamental group or winding-number theorem is needed.

On each compact smooth level, h is nonconstant on each component because its fibres are finite. Its critical points are therefore isolated and finite; there are none on the boundary collar. Away from their finitely many values, properness and the implicit function theorem give a covering. The punctured base disc is path connected, and its regular fibres have the same cardinality as those near its outer circle, namely S.

3.5 Local factors, without a generic-Morse projection

Fix a critical value c. Its finitely many preimages have disjoint normalization neighbourhoods and distinct ζ coordinates, by injectivity of $\boldsymbol{n}$. There are only finitely many such fibres. Rotate the

complementary ζ coordinate by a fixed unit complex number avoiding the finitely many directions for which two distinct values in one of those fibres have the same real part. Choose small ζ discs whose real projections are disjoint within each such fibre. Thus the sheets of each local cluster occupy consecutive positions in their real-coordinate order. Choose a sufficiently small disc about c so that each moving cluster stays in its own disc. The remaining sheets stay away from those discs.

At a singular point Q_j, the supplied branch-word construction in the transverse coordinates (h, ζ) gives the positive event word β_j on a_j sheets, with the cycle, generator-occurrence, and length properties stated in the question. We use that explicit event-word construction, not only a statement about the knot type of its closure.

At a smooth critical point, a normalization parameter can be chosen so that $h - c = t^d$ for some $d \geq 2$: take a holomorphic dth root of the unit in its leading expansion. Since the plane curve is smooth there, $\zeta(t)$ has nonzero derivative and is injective on a smaller parameter disc. Replace t^d locally by

$$p_u(t) = t^d - u\varepsilon t, \qquad 0 \leq u \leq 1,$$

where $\varepsilon \neq 0$ is sufficiently small. On a fixed sufficiently small circle $|h - c| = r$, all d roots remain in that parameter disc and remain distinct throughout the family. To verify this explicitly, the critical points satisfy $dt^{d-1} = u\varepsilon$, and their critical values are $(1 - d)t^d$, whose absolute values are less than r when ε is small. Thus no critical value reaches the boundary circle; continuity of roots and injectivity of $\zeta(t)$ give a continuous family of configurations of distinct points there.

For $u = 1$, the $d-1$ critical points are simple and their critical values are distinct: multiplication by ε/d identifies the distinct roots of $dt^{d-1} = \varepsilon$ with their dth powers. Near each of them the projection is, after a holomorphic coordinate change, $t \mapsto t^2$. Its two roots make one counterclockwise half twist. Cutting its local base disc along paths to these $d - 1$ values therefore expresses its boundary configuration as a product of $d - 1$ conjugated positive single-generator events. The preceding boundary family identifies this factorization with the original ordinary ramification. The local conjugating words are simply concatenated with its transport word in the global base.

If several clusters have the same original critical value, their motions take place in disjoint ζ discs. They can be performed consecutively instead of simultaneously: replace their common angular parameter by nondecreasing functions which run one cluster at a time. Linear interpolation of these parameter functions is an explicit homotopy of configurations, because points in different discs never meet. Thus the same factorization separates all the singular-cluster factors and all the ordinary positive single-generator factors, without assuming distinct original critical values.

3.6 Why the configuration homotopies give finite word relations

We record the elementary finite-event verification used above and below. A continuous family of configurations of S distinct complex points on a compact parameter rectangle has a positive minimum pairwise separation. Its labels can be continued on the rectangle: cover the image locally by disjoint root discs and continue labels across a fine triangulation; consistency around each small triangle follows because each root stays in its own disc. Triangle deletion gives consistency on the whole rectangle.

For a transport path which has not already been assigned an event word, use a generic piecewise-affine approximation within these root discs, keeping its endpoint configurations fixed. The finitely many endpoint configurations can be chosen with distinct real coordinates, by a small rotation and by moving the cut endpoints on their regular circles. The resulting path has finitely many transverse events and defines its word. Thus no finiteness or transversality assertion about an arbitrary unperturbed continuous transport path is needed.

Approximate the labelled coordinates by piecewise-affine functions on a sufficiently fine

triangulation, keeping the error smaller than one quarter of the minimum separation. The approximation, and the straight homotopy to it, still have distinct points. On boundary paths already used to define words, choose vertices between their finitely many transverse events; sufficiently small changes preserve their words. The vertex coordinates may now be chosen generically within those open error bounds. Only finitely many polynomial conditions are to be avoided: a real-coordinate tie at a vertex, coincident zero lines in a triangle, four equal real coordinates, three independent pair-equalities, an event intersection on a mesh edge, or equality of the movie-parameter coordinates of two unrelated exceptional events. Each is a proper polynomial condition in the free vertex coordinates. For example, three independent affine pair-differences have a common zero only if their three coefficient rows have zero determinant; its determinant is not identically zero. Structural equalities involving three indices are retained, since the three pair-differences then satisfy their necessary linear relation. Moving mesh vertices slightly, if needed, also avoids vertical interior edges. Finitely many proper polynomial zero sets cannot fill any open choice of the vertex coordinates.

In every triangle a pair-equality $\mathrm{Re}(z_i - z_j) = 0$ is now a straight segment. These segments form a finite arrangement. As a vertical movie-parameter slice moves, its event word changes only in the following ways. First, at a local turn of one pair-equality arc along a mesh edge, two crossings of the same pair are born or disappear; their signs are opposite because the real difference crosses zero in opposite directions while the imaginary difference keeps its sign. Second, two independent pair-equality arcs cross, interchanging events on disjoint adjacent pairs, which is far commutation. Third, three real coordinates coincide. Their imaginary coordinates remain distinct, and their six possible height orders give the braid relation, its inverse, or one of the four mixed-sign forms

$$aba^{-1} = b^{-1}ab, \quad ab^{-1}a^{-1} = b^{-1}a^{-1}b, \quad a^{-1}ba = bab^{-1}, \quad a^{-1}b^{-1}a = ba^{-1}b^{-1}.$$

Each displayed relation follows from $aba = bab$ by multiplication and cancellation. The arrangement has finitely many segments, turns, and intersections, so this procedure supplies a finite list of these word operations. It neither invokes Reidemeister's theorem nor identifies a configuration-space fundamental group with an Artin group.

This argument first applies to a homotopy of loops with fixed basepoint configuration, for which no event reaches the two endpoint edges of the parameter rectangle. In a loop homotopy L_s with moving basepoint $b(s)$, replace L_s by the based loop

$$b|_{[0,s]} * L_s * \overline{b|_{[0,s]}}.$$

These loops form a continuous homotopy with basepoint $b(0)$. The preceding argument then compares the initial event word with the conjugate of the final event word by the transport word of b. Equivalently, keeping an unbased loop cut records a crossing through that cut as a cyclic word rotation. This treats the boundary case as well as every interior event of the arrangement.

3.7 Global factorization, exponent count, and transitivity

Choose polygonal cuts from a regular boundary basepoint to disjoint small discs about the distinct critical values. The cuts are disjoint except at their basepoint and have a specified cyclic order there. The cut disc with the small discs removed has a finite triangulation. Removing its triangles successively deforms its outer boundary to the ordered product of path–small-circle–reverse-path loops. The roots are distinct throughout. Apply the finite-event verification above and the local factorizations just proved. It gives an equality by explicit finite event relations of the form

$$w_{\mathrm{out}} \sim \prod_{l=1}^{N+B} \gamma_l b_l \gamma_l^{-1},$$

where the N singular factors are the positive words β_j and the other B factors are positive single generators. Here B is the sum of the ordinary ramification orders $d-1$; it need not be the number of original ordinary critical points.

Over $|h| = \rho$ the source sheets vary by the implicit function theorem without collision as v moves from zero to the chosen level, and $\zeta \circ \boldsymbol{n}$ separates them by injectivity. Thus their outer configurations form a homotopy of closed S-braids. If its basepoint configuration moves, include its transport path; the resulting word comparison is a conjugacy, again furnished by the finite-event verification. The central branch construction supplied in the question then compares this outer word with $\mathbf{d}_S\mathbf{w}$. All the relations and conjugacies preserve exponent sums. Therefore

$$\begin{aligned} B &= 2\delta(W) + S - b - \sum_j (2\delta_j + a_j - 1) \\ &= S - \sum_j a_j + N - 1 + \mu(V). \end{aligned}$$

This computation uses only the supplied branch lengths and delta identities; no argument principle or Riemann–Hurwitz calculation is needed here.

Finally let $\mathcal{V}$ be the finite set of critical values of the original finite projection. Remove the *entire* finite set $h^{-1}(\mathcal{V})$ from the connected smooth normalization E_s, not just its ramification points. A connected real surface remains path connected after finitely many interior points are removed: take a finite chartwise path and detour around those points in small coordinate discs. A path in this complement between two points of one regular fibre projects to a loop avoiding $\mathcal{V}$.

The monodromy of that loop is generated by the small critical-value loops. Explicitly, approximate it polygonally away from the punctures, make it transverse to the chosen cuts, and contract each portion in the cut region by deleting triangles. Its finitely many crossings of the cuts record a word in the path–circle–reverse-path loops. This gives the required expression without a fundamental-group presentation. Consequently the permutations of the original critical-value loops act transitively on the S sheets. Every such permutation is a product of the permutations of the separated factors above, including the $d-1$ factors replacing an ordinary higher ramification. Hence the permutations of all $N+B$ displayed factors also generate a transitive group, as required.

□

4 A rectangle proof of the native O-saddle pairing

Lemma 5 (Native O calculus and incidence pairing; new combinatorial)**.**

Proposed name: `nativeOSaddleIncidence`.

For every labelled O diagram of size at least two, the differentiated rectangle maps Φ_i below are polynomial chain maps of degree $+1$, annihilating the canonical class on homology. An adjacent O exchange has the explicit diagonal degree--1 maps F, F' below, with shifts $(0, -1/2)$. For either of the two exchanged labels i, $K_i = \Phi_i F + F\Phi_i$ is a polynomial degree-zero homotopy equivalence, preserves the canonical class, and intertwines all marking derivatives on homology. For any sequence of O exchanges, with intervening degree-zero polynomial canonical-class maps intertwining the colour derivatives, the product of as many colour derivatives as O exchanges has the determinant action (13). In particular a rooted-tree incidence matrix gives coefficient one.

Proof. All coefficients in this section have characteristic two. Label the n O markings and put $R = \mathbb{F}_2[U_1, \dots, U_n]$. The full multivariable grid differential is

$$D_G x = \sum_y \sum_{r \in \mathrm{Rect}^\circ(x,y)} \left(\prod_{h=1}^{n} U_h^{n_h(r)} \right) y, \qquad n_h(r) = \#(r \cap O_h) \in \{0, 1\}.$$

Its square is zero: nonannular index-two domains have two decompositions into empty rectangles; the remaining domains are the horizontal and vertical annuli, and each O_h occurs in precisely one annulus of each kind, giving $U_h + U_h = 0$. This argument counts rectangles containing X markings. Write d_G for the specialization $U_h = T$ for every h.

4.1 Derivative maps and a rectangle homotopy

Define

$$\Phi_h^G = \left.\frac{\partial D_G}{\partial U_h}\right|_{U_1=\cdots=U_n=T}, \qquad \Phi_h^G(x) = \sum_{y,r} n_h(r) T^{n_O(r)-1} y.$$

Every nonzero summand has $n_O(r) \geq 1$, so this is a polynomial map, including when the original complex has been localized in T. It has Maslov degree $+1$ and first filtration shift zero. Since

$$A(y) - n_O(r) + 1 - A(x) = 1 - n_X(r),$$

its second filtration shift is at most 1. If O_h and an X marking occupy the same square, the second shift is at most zero.

Differentiating $D_G^2 = 0$ proves

$$d_G \Phi_h^G + \Phi_h^G d_G = 0.$$

The differential is linear in each individual U_h, so its coefficient of U_h^2 in D_G^2 proves $(\Phi_h^G)^2 = 0$ strictly. For $h \neq k$, twice differentiating proves

$$\Phi_h^G \Phi_k^G + \Phi_k^G \Phi_h^G = d_G B_{hk} + B_{hk} d_G, \qquad B_{hk} = \left.\frac{\partial^2 D_G}{\partial U_h \partial U_k}\right|_{U=T}.$$

Thus the induced derivative operators commute on homology.

Let p lie in the interior of a vertical grid circle, away from the horizontal grid circles, and suppose the two columns adjacent to this circle contain O_i and O_j. For a rectangle r let $a_p(r)$ be one if its vertical boundary passes through p, and zero otherwise. Define the multivariable map

$$\mathcal{A}_p^G(x) = \sum_{y,r} a_p(r) \prod_h U_h^{n_h(r)} y.$$

Then

$$D_G \mathcal{A}_p^G + \mathcal{A}_p^G D_G = (U_i + U_j)\,\mathrm{id}\,. \tag{5}$$

Here is the entire additional check beyond the usual proof that $D_G^2 = 0$. For a decomposition $r_1 * r_2$ the contribution to the left side is weighted by $a_p(r_1) + a_p(r_2)$. This is the multiplicity at p, modulo two, of the vertical boundary of the composite domain, and hence is independent of its decomposition. The two decompositions of every nonannular domain therefore cancel with the same weight. The vertical boundary of a horizontal annulus is zero. A vertical annulus has weight one precisely when it is one of the two annuli adjacent to the circle containing p. These two annuli contain O_i and O_j, respectively. Each has the usual unique empty-rectangle decomposition at each state, giving exactly the right side of (5).

Let $A_p^G = \mathcal{A}_p^G|_{U=T}$. Differentiating (5) in U_i and then specializing gives

$$\Phi_i^G A_p^G + A_p^G \Phi_i^G = \mathrm{id} + d_G B_i^G + B_i^G d_G, \qquad B_i^G = \left.\frac{\partial \mathcal{A}_p^G}{\partial U_i}\right|_{U=T}. \tag{6}$$

The same identity holds with j in place of i.

4.2 The O switch and its two detecting operators

Consider adjacent nonwrapping columns $a, a+1$. In G suppose

$$O_i = (a+\tfrac{1}{2}, r+\tfrac{1}{2}), \qquad O_j = (a+\tfrac{3}{2}, s+\tfrac{1}{2}), \qquad r < s.$$

In H interchange these two O markings, preserving their labels and their rows. All other O markings and all X markings stay fixed. Set

$$e(x) = \mathbf{1}_{\{r<x(a+1)\le s\}}, \qquad F(x) = T^{e(x)}x, \qquad F'(x) = T^{1-e(x)}x.$$

The elementary J-function calculation gives

$$M_O^H(x) - M_O^G(x) = 2e(x) - 1.$$

The rectangle Maslov formula consequently gives, for every empty rectangle $q : x \longrightarrow y$,

$$n_O^H(q) - n_O^G(q) = e(y) - e(x). \tag{7}$$

It follows term by term that F and F' are chain maps of Maslov degree -1, with shifts $(0, -\frac{1}{2})$, and

$$F'F = T\,\mathrm{id}, \qquad FF' = T\,\mathrm{id}\,.$$

The reverse local pattern is covered by interchanging F, F' and G, H. A cyclic permutation of columns reduces a wrapping column pair to this case.

Define the degree-zero maps

$$K_i = \Phi_i^H F + F\Phi_i^G, \qquad K_i' = \Phi_i^G F' + F'\Phi_i^H.$$

Both preserve the polynomial subcomplex, and both have shifts $(0, \frac{1}{2})$. Take $p = (a+1, r+\frac{1}{2})$. A rectangle contains exactly one of O_i^G, O_i^H precisely when its vertical boundary passes through p. This is because the two points lie in adjacent squares on the same horizontal level, and the unique grid circle between them is $x = a+1$. Using (7) to compare every individual rectangle therefore proves the exact identities

$$F'\Phi_i^H F = T\Phi_i^G + A_p^G, \qquad F\Phi_i^G F' = T\Phi_i^H + A_p^H. \tag{8}$$

For example, the coefficient of a rectangle $q : x \to y$ on the left of the first identity is

$$n_i^H(q) T^{e(x)+n_O^H(q)-1+1-e(y)} = n_i^H(q) T^{n_O^G(q)}.$$

The right side has coefficient $(n_i^G(q) + a_p(q))T^{n_O^G(q)}$, which is the same.

Expanding and using $(\Phi_i^G)^2 = (\Phi_i^H)^2 = 0$ gives

$$\begin{aligned} K_i'K_i &= \Phi_i^G(F'\Phi_i^H F) + (F'\Phi_i^H F)\Phi_i^G \\ &= \Phi_i^G A_p^G + A_p^G\Phi_i^G = \mathrm{id} + d_G B_i^G + B_i^G d_G. \end{aligned}$$

The identical calculation in H gives $K_iK_i' = \mathrm{id} + d_H B_i^H + B_i^H d_H$. Thus K_i is a polynomial-preserving, degree-zero chain homotopy equivalence, with the displayed explicit inverse and homotopies.

For later use, the induced K_i intertwines every labelled derivative operator. To prove this, first work over the ring in which only $U_i = U_j = u$ has been imposed. The same rectangle calculation makes $F = u^e\,\mathrm{id}$ a chain map over this ring. Differentiation in u proves

$$K_i + K_j = d_H J + J d_G, \qquad J(x) = e(x)x.$$

Differentiation in any other U_k proves the strict identity $\Phi_k^H F = F\Phi_k^G$. Now $\Phi_i^H K_i = K_i\Phi_i^G$ follows strictly from the square-zero identity. For j it follows on homology by replacing K_i with K_j. For any other k, it follows from the strict identity for F and the commutation, on homology, of the derivative maps. In particular,

$$(\Phi_k^H)_*(K_i)_* = (K_i)_*(\Phi_k^G)_* \quad \text{for every label } k. \tag{9}$$

4.3 Why every derivative annihilates the canonical class

We first prove the slightly stronger fact

$$\operatorname{im}(H_m(P(G)) \longrightarrow H_m(C(G))) = 0 \qquad (m > 0). \tag{10}$$

This uses the explicitly proved K_i equivalences and a finite permutation inequality; it uses no homology computation or grid isotopy theorem. The differential, Maslov grading, and polynomial subcomplex ignore X. Work temporarily with O-only grids and bubble-sort the O permutation to $O(c) = n - 1 - c$ by adjacent column exchanges. Each exchange has the degree-zero polynomial-preserving equivalence K_i constructed above. Its proof did not use (10) or any assertion about the canonical class. Each K_i is injective on Laurent homology and maps the polynomial image into the polynomial image. It therefore suffices to verify (10) for antidiagonal O. If ordinary grid notation is desired throughout, choose an arbitrary disjoint X permutation at each stage; neither the complexes in this argument nor their first filtrations change.

For a state x, put $\pi(c) = n - 1 - x(c)$, let $F(\pi) = \sum_c |\pi(c) - c|$, let $D(\pi) = \#\{c : \pi(c) < c\}$, and write $\operatorname{inv}(\pi)$ for its inversion count. Directly from J,

$$M_O(x) = \operatorname{inv}(\pi) - \sum_c |\pi(c) - c + 1| + 1 = \operatorname{inv}(\pi) - F(\pi) - n + 2D(\pi) + 1.$$

If $c(\pi)$ denotes the number of cycles, then

$$\operatorname{inv}(\pi) + n - c(\pi) \le F(\pi), \qquad D(\pi) \le n - c(\pi).$$

For completeness, prove the first inequality by repeatedly taking the least nonfixed index i, putting $j = \pi^{-1}(i) > i$, and interchanging the values at i, j. Write $v = \pi(i) > i$. This decreases the footrule by $2\min(v - i, j - i)$ and decreases the inversion count by

$$1 + 2\#\{k : i < k < j,\ i < \pi(k) < v\} \le 2\min(v - i, j - i) - 1.$$

It therefore decreases $F - \operatorname{inv}$ by at least one. The transposition splits off the fixed point i from its cycle, and hence exactly $n - c(\pi)$ such steps produce the identity. This proves the first inequality. The second follows because each cycle of length q has at most $q - 1$ deficient indices. Consequently

$$M_O(x) \le -(n - c(\pi)) - n + 2D(\pi) + 1 \le D(\pi) - n + 1 \le 0.$$

Every polynomial monomial thus has nonpositive Maslov degree. This proves (10). Equality $M_O = 0$ forces $D(\pi) = n - 1$ and hence $\pi = (n - 1, 0, 1, \dots, n - 2)$, giving the unique state $\theta = (0, n - 1, n - 2, \dots, 1)$. Thus the degree-zero polynomial chain group is the one-dimensional space generated by θ. The supplied canonical-class theorem says that its degree-zero homology has nonzero image in Laurent homology. It follows at once that θ is a cycle with nonzero Laurent class, and this class is the canonical class in the normal form.

Since Φ_i^G is polynomial preserving of degree $+1$, applying it to a polynomial degree-zero representative of the canonical class and using (10) gives

$$(\Phi_i^G)_* \operatorname{can}(G) = 0.$$

Because K_i has degree zero, preserves the polynomial subcomplex, and is an isomorphism on homology, Lemma N gives $(K_i)_* \operatorname{can}(G) = \operatorname{can}(H)$. We have therefore proved the explicit O-switch pairing identities

$$(\Phi_i^H)_* F_* \operatorname{can}(G) = (\Phi_j^H)_* F_* \operatorname{can}(G) = \operatorname{can}(H), \qquad (\Phi_k^H)_* F_* \operatorname{can}(G) = 0 \quad (k \notin \{i, j\}). \tag{11}$$

4.4 Component colours and the determinant formula

The fresh-label hypothesis can be removed. Assign each O marking a colour in a fixed finite set. At a given stage put

$$\Psi_c = \sum_{\mathrm{col}(i)=c} \Phi_i.$$

These are degree-one chain maps, polynomial preserving, whose induced operators commute and square to zero. Indeed, the squares of the individual Φ_i vanish, and every mixed summand in Ψ_c^2 is an anticommutator, already proved null-homotopic. Each Ψ_c annihilates the canonical class by the positive-degree polynomial vanishing theorem.

Suppose an O switch F_e exchanges labels i_e, j_e. The colour is attached to the labelled marking and moves with it. Set

$$b_{c,e} = \mathbf{1}_{\{\mathrm{col}(i_e)=c\}} + \mathbf{1}_{\{\mathrm{col}(j_e)=c\}} \ \in \mathbb{F}_2.$$

On homology the identities already proved give

$$(\Psi_c^{\mathrm{after}})_*(F_e)_* = (F_e)_*(\Psi_c^{\mathrm{before}})_* + b_{c,e}(K_e)_*. \tag{12}$$

The same K_e can be used for either endpoint of the edge since $K_{i_e} \simeq K_{j_e}$. Moreover K_e intertwines every Ψ_c.

Let $F_1, \dots, F_d$ be any sequence of d O switches, and choose d colours $c_1, \dots, c_d$. Let $B = (b_{c_r,e})_{1\le r,e\le d}$. Then

$$(\Psi_{c_1}^{\mathrm{final}})_* \cdots (\Psi_{c_d}^{\mathrm{final}})_*(F_d)_* \cdots (F_1)_* \,\mathrm{can}(G_0) = \det(B)\ \mathrm{can}(G_d). \tag{13}$$

To prove the formula, move the derivatives successively to the right using (12). At a position still occupied by F_e a derivative either passes through or consumes that position, replacing F_e by $b_{c,e}K_e$. A later derivative passes through a K_e without any extra term, so a position can be consumed at most once. Every term in which a derivative reaches the initial canonical class vanishes. Since there are exactly d derivatives and d positions, each surviving term consumes every position exactly once. Such terms are indexed by bijections between the derivatives and the positions. Their coefficients sum to the permanent of B, which equals its determinant in characteristic two. In every such term the operator product is exactly $K_d \cdots K_1$, in its original order; no commutativity of a K with an F is being used. That product sends the initial canonical class to the final one. This proves the formula.

There may also be arbitrary intervening degree-zero, polynomial-preserving canonical-class maps P which intertwine the colour derivatives. The same expansion passes derivatives through each P and proves the identical formula with the P's left in their original positions.

Here is the precise coefficient rule that gives colour naturality for native moves. Replace the independent marking variables by colour variables V_c, namely $U_i = V_{\mathrm{col}(i)}$, and denote the resulting differential by D^{col}. For any chain map P^{col} defined over this polynomial ring, differentiation of

$$D_H^{\mathrm{col}} P^{\mathrm{col}} = P^{\mathrm{col}} D_G^{\mathrm{col}}$$

in V_c, followed by $V_c = T$, proves

$$\Psi_c^H P + P\Psi_c^G = d_H P_c + P_c d_G, \qquad P_c = \left.\frac{\partial P^{\mathrm{col}}}{\partial V_c}\right|_{V=T}.$$

For a stabilization the new O marking must inherit the colour of the old O marking whose variable it is identified with; a destabilization uses the reverse rule. Under that rule the native polygon formulas lift to the colour ring: paired domains have the same complete marking monomial, and the exceptional stabilization annulus terms have the two equal colour variables. This explains

exactly why differentiating by one arbitrarily chosen representative marking is insufficient, whereas the full colour derivative is natural under these moves.

Finally, suppose the graph whose vertices are the colours and whose edges are the O switches is a tree on $d+1$ vertices. Choose one root colour and choose all the other d colours in (13). Then B is its incidence matrix with the root row removed, so $\det(B) = 1$. A proof requiring no graph linear algebra is induction on d: a nonroot leaf exists, its row has exactly one entry equal to one, and expansion along that row removes that leaf and its incident edge. The empty tree has determinant one. The same argument applies component by component to a forest with one root per component.

To apply this to deaths, colour every marking of each final isolated unknot by the colour of that component. Ordinary moves reducing that component to a doubly marked square preserve its colour derivative by the preceding rule. At the doubly marked square the derivative has second filtration shift zero. The degree-one death can be written $R\Psi_c$, where R is the degree-zero top projection and has second shift $+\frac{1}{2}$. The derivative for a different dying component commutes past R on homology, since that projection changes no marking of the other colour. All deaths therefore give a product of the component derivatives followed by the degree-zero top projections. If their switch incidence graph is the rooted forest just described, (13) proves that the complete map carries the canonical class to the canonical class.

□

5 Auxiliary contractions and four-mark normalization

In this section only, an *auxiliary marking pair* is a pair of permutations (O, X) for which some values $O(c) = X(c)$ are allowed. This does not enlarge the class of native endpoints. Its chain module, differential, and two grading functions are given by exactly the formulas in the question. The complex and its polynomial image depend only on O. Consequently its canonical class is defined by identifying this complex with the native complex of any valid pair having the same O. Such a valid pair exists in every size at least two; for example, take $X(c) = O(c+1)$ cyclically. The resulting class is independent of this choice, because these are literal identifications of the chain complex and of its polynomial subcomplex.

The derivative Φ_i is the explicitly defined rectangle map of Lemma 5. At a coincident O/X square it has degree $+1$ and shifts $(0, 0)$, by $A(y) - n_O(q) + 1 - A(x) = 1 - n_X(q) \le 0$.

5.1 Exact local coordinates

Lemma 6 (Local auxiliary contraction; new combinatorial)**.**

Proposed name: `nativeAuxiliaryContraction`.

Let G be a valid grid of size $N \ge 2$ and choose an X square (c, r), so $G.X(c) = r$. Insert a column immediately to the right of column c and a row immediately above row r. Embed the old indices by

$$\iota_c(a) = \begin{cases} a & a \le c, \\ a+1 & a > c, \end{cases} \qquad \iota_r(b) = \begin{cases} b & b \le r, \\ b+1 & b > r. \end{cases}$$

Let B be the auxiliary pair retaining all embedded old markings and having an additional O and X together in square $(c+1, r+1)$. Label this new O marking i. There is an explicitly specified chain map

$$D_B : C(B) \longrightarrow C(G)$$

of degree $+1$ and shifts $(0, \frac{1}{2})$. There is also a degree-zero map $\rho_B : C(B) \to C(G)$ with those

shifts and with $\rho_{B*}\,\mathrm{can}(B) = \mathrm{can}(G)$*, such that*

$$D_B = \rho_B \Phi_i.$$

Proof. Let A have the same O markings as B, and replace the two X markings $(c, r), (c+1, r+1)$ by $(c, r+1), (c+1, r)$. In the indicated 2×2 block A therefore has

	c	$c+1$
$r+1$	X	O_i
r	$\varnothing$	X

and all its other markings are the embedded markings of G. This is exactly an X stabilization of G, with the southwest square empty. It is valid: the old O in column c was not in row r, so after insertion it occupies neither of the displayed X squares. All remaining validity conditions follow from those of G.

Use the explicit type-R destabilization count $R_A : C(A) \to C(G)$ defined below. Section 6 proves directly that this particular formula has degree zero, shifts $(0, 0)$, and preserves the canonical class. The identity of the underlying O-complexes defines $I : C(B) \to C(A)$. The usual adjacent-X grading calculation depends only on the X permutations, not on disjointness from O, and therefore applies here: $M_X^A(x) - M_X^B(x) \in \{-1, 1\}$ for every state. Hence I has shifts $(0, 1/2)$. Because it is a literal O-complex and polynomial-subcomplex identity, it preserves the canonical class.

Set

$$\rho_B = R_A I, \qquad D_B = R_A I \Phi_i.$$

The preceding derivative calculation applies to B, since O_i and its X coincide. Thus Φ_i has degree $+1$ and shifts $(0, 0)$, which proves every stated chain, degree, and filtration property. The canonical-class identity for ρ_B follows from the identities for I and R_A. □

Corollary 7 (Paired contraction; new combinatorial)**.**

Proposed name: `pairedAuxiliaryContraction`.

Suppose an O-switch map $F : C(K) \to C(B)$ *has degree* -1*, shifts* $(0, -1/2)$*, and satisfies*

$$\Phi_{i*} F_* \,\mathrm{can}(K) = \mathrm{can}(B).$$

Then $D_B F$ *has degree zero, shifts* $(0, 0)$*, and sends* $\mathrm{can}(K)$ *to* $\mathrm{can}(G)$*.*

Proof. Compose the maps and their grading bounds, and use $D_B = \rho_B \Phi_i$ and the canonical-class identity for ρ_B. □

5.2 Labelled naturality and the persistent root variable

For global compositions, choose a label 0 which survives all the contractions. Place the auxiliary square in Lemma 6 next to the X in the row of O_0. The O marking traditionally paired with O_i in this stabilization is then exactly O_0. Use the coefficient ring

$$\mathbb{F}_2[T, U_j : j \neq 0, i], \qquad U_0 = U_i = T,$$

for its top destabilization map. The explicit polynomial representative of that map is the following. Put $p = (c+1, r+1)$ at the crossing of the newly inserted circles, and let $e(y)$ delete p from a state y containing it, followed by removal of the new row and column. A domain $q : x \to y$ is an array of integers $q_{a,b}$ on the grid squares, with cyclic indices, satisfying at every intersection (a, b) the boundary equation

$$q_{a,b} + q_{a-1,b-1} - q_{a-1,b} - q_{a,b-1} = \mathbf{1}_{\{x(a)=b\}} - \mathbf{1}_{\{y(a)=b\}}.$$

This is the northeast-plus-southwest minus northwest-minus-southeast corner convention used for the empty rectangles. Count nonnegative domains from x to y of type R: at every point of $(x \cup y) \setminus \{p\}$ at least three adjacent multiplicities vanish; at p three multiplicities are k and the southeast multiplicity is $k+1$; and $y \setminus x$ has $2k+2$ points. A domain contributes

$$\left(\prod_{j \neq i} U_j^{n_{O_j}(q)}\right) e(y).$$

The set of domains is finite: the number $2k+2$ is at most the grid size, and the zero-multiplicity conditions determine a domain of each boundary type uniquely. There is no factor U_i in this weight. This is the usual top projection representative of the native X-destabilization map.

The verification of its chain equation retains these separate variables. Its cancellations have two forms. In the first, a rectangle and a type-R domain are cut in the other order, with the same total domain; the marking monomials are therefore identical. In the second, the alternative domains differ by one of the thin annuli in the inserted row or column. These annuli contain O_i or the O marking in the row of the lower X, namely O_0, and no other O markings. Their coefficient contributions agree after $U_i = U_0 = T$. No identification of any other U_j is involved. Equivalently, before that identification the only uncancelled term is $(U_i + U_0)$ times the complementary type-L domain count. This establishes the required chain equation over the displayed ring, rather than merely after setting every variable to T.

Consequently, for each surviving label $j \neq 0, i$, differentiating the chain equation and then setting all variables to T yields

$$\Phi_j^G R_A + R_A \Phi_j^A = \partial R_{A,j} + R_{A,j}\partial, \qquad R_{A,j} = \left.\frac{\partial R_A}{\partial U_j}\right|_{U=T}.$$

The identity I changes no O label or differential, so the same homotopy relation holds for ρ_B. This is exactly the naturality needed to move the derivative operators of the remaining contractions past ρ_B. The root label is never differentiated. If instead a surviving label is used as the variable paired with U_i, differentiation also produces $R_A \Phi_i$; choosing the persistent root as above avoids that extra term.

5.3 Moving and removing an arbitrary separated auxiliary component

The preceding construction also works when G has other doubly-marked squares, provided the chosen X square (c, r) is not itself doubly marked. The type-R formula depends on the O markings and the position of the stabilization, but not on any other X marking. Its chain equation and its independent canonical-class proof in Section 6 hold for all O diagrams. The Alexander estimate there is the negative of a sum of nonnegative X multiplicities. It therefore remains valid when some of those X markings coincide with O markings. The derivative estimate at the distinguished doubled square and the adjacent-X-exchange estimate likewise remain valid. Hence Lemma 6 applies in this more general situation.

For the same reason, commutation maps extend to auxiliary pairs whenever their marking segments are noninterleaving. More explicitly, use the usual empty-pentagon count, putting the O-multiplicity monomial on every pentagon. Its differential and homotopy-inverse verifications pair domains with identical O multiplicities, exactly as for a native commutation. None of these verifications uses disjointness of the X and O markings. The grading comparison for a pentagon is unchanged and its Alexander loss is the total nonnegative X multiplicity. Thus both directions have degree zero and shifts $(0, 0)$, and the homotopy equivalence preserves the polynomial image and hence the canonical class. Keeping separate O variables in these domain identities also gives labelled derivative naturality by differentiation.

Here is an explicit check of the degenerate segment used below. Suppose the doubled square is in the left of the two adjacent columns, whose horizontal coordinates are $(a, a+1)$ and $(a+1, a+2)$, and its marking height is $v + 1/2$. Choose a small open interval J about $v + 1/2$, contained in that row and containing no marking height of the other column. Replace the intervening vertical circle $x = a + 1$ by the graph $x = f(y)$, periodic in y, with $f = a + 1/4$ on a smaller interval about $v + 1/2$, $f = a + 7/4$ outside J, and exactly one monotone transition at each end of J. This graph avoids every marking, meets each horizontal grid circle once, and intersects $x = a + 1$ exactly twice. It puts the doubled marking in the right region and both markings of the other column in the left region, and hence gives precisely the commuted diagram after straightening. Reflection gives the construction for a doubled right column. Interchange the coordinates for a row move.

To check every grading and cancellation at the coinciding punctures, choose a disk around the doubled marking disjoint from all grid circles and from this new graph. Temporarily separate O and X by arbitrarily small distinct displacements inside that disk. The small segment between the displaced heights still lies in J, so the same commutation curves work. Each rectangle, pentagon, and hexagon domain has constant multiplicity on the disk; therefore its O and X multiplicities are unchanged by bringing the two points back together. The pairwise orders of the X markings and their orders relative to every state coordinate are unchanged as well, so the J-grading comparisons are unchanged. The rectangle–pentagon and pentagon–hexagon cancellation proofs use only these incidences and multiplicities. They thus verify the maps, homotopies, and filtration bounds also at the coinciding marking. This is a direct verification of the polygon formulas on auxiliary pairs, not an appeal to a topological invariance theorem for them.

Lemma 8 (Singleton transport; new combinatorial)**.**

Proposed name: `auxiliarySingletonTransport`.

Let E be an auxiliary pair of size $n \geq 3$ with $E.O(u) = E.X(u) = v$, and label this doubled marking i. Delete its column and row and call the resulting auxiliary pair G, of size $n - 1$. Choose a surviving root marking whose row r in G is not doubly marked, and put $c = G.X^{-1}(r)$. There is an explicit finite sequence of noninterleaving commutations taking E to $B(G, c, r)$, with all labels retained. In particular there is a degree-$+1$ map

$$\mathcal{E}_i : C(E) \longrightarrow C(G)$$

with shifts $(0, 1/2)$, and a degree-zero canonical-class-preserving map $\rho_i : C(E) \to C(G)$ with the same shifts, such that

$$(\mathcal{E}_i)_* = (\rho_i)_*(\Phi_i)_*.$$

The maps intertwine every surviving derivative except possibly that of the root; the root derivative is not needed.

Proof. For clarity, deleting the dummy means the exact formulas

$$G.Y(a - \mathbf{1}_{a>u}) = E.Y(a) - \mathbf{1}_{E.Y(a)>v} \quad (a \neq u, \quad Y \in \{O, X\}).$$

The desired insertion indices are $u_* = c + 1$ and $v_* = r + 1$. If $u < u_*$, commute the dummy column successively across the pairs $(u, u + 1), (u + 1, u + 2), \ldots, (u_* - 1, u_*)$; if $u > u_*$, use the reverse list $(u - 1, u), (u - 2, u - 1), \ldots, (u_*, u_* + 1)$. Use no column moves if $u = u_*$. Then apply the analogous list of row commutations to move the dummy row from v to v_*. There are exactly $|u - u_*| + |v - v_*|$ commutations.

Every one is noninterleaving. The dummy column has both endpoints at the same row, and no other column has either marking in that row, because the dummy already uses both the unique O and the unique X of that row. A singleton segment cannot interleave with a segment whose endpoints are distinct from that singleton. The same argument applies to the dummy row. It applies also when the adjacent column or row belongs to another doubled marking. These moves

never change the relative order of the retained rows or the retained columns. Therefore, after deletion of the dummy, their result is still exactly G with its original indexing. With insertion positions $c+1, r+1$, the resulting pair is exactly $B(G, c, r)$, as asserted.

Let $P : C(E) \to C(B(G, c, r))$ be the composition of the commutation maps. Set

$$\rho_i = \rho_B P, \qquad \mathcal{E}_i = D_B P = \rho_B \Phi_i^B P.$$

All the degree, filtration, and canonical-class assertions follow from the preceding lemmas. Differentiating the fully labelled commutation identities proves $\Phi_i^B P \simeq P \Phi_i^E$, giving the displayed identity on homology. Naturality for the other surviving labels follows from the same commutation identities and from the root-variable destabilization identity proved above. □

Consequently, after any prior algebraic or geometric cancellation has left a list of doubled singleton components, their removal is an explicit operation on the marking permutations: apply the adjacent transposition lists in Lemma 8 and then the displayed contraction. There is no additional link-equivalence or component-placement assertion at this stage. The remaining hypothesis for a global application is that its previous construction has indeed produced these singleton components and a surviving root row with distinct O and X markings.

5.4 The component-colour version

The same statements hold when several O markings have one common component colour. Assign a variable V_α to each colour and substitute $U_j = V_{\mathrm{colour}(j)}$ in the multivariable differential. Its derivative, after collapse to T, is

$$\Phi_\alpha = \sum_{j:\mathrm{colour}(j)=\alpha} \Phi_j.$$

This follows by the polynomial chain rule, including terms whose rectangle contains several markings of that colour. Thus derivative naturality follows directly from the same chain-map identities, with no requirement that a colour have only one O marking. In an ordinary stabilization, give its new O the colour of the component on which the stabilization is performed. The two variables identified by its top projection then already coincide.

In a dummy contraction, let δ be the colour of the deleted singleton and α_0 the persistent root colour. The top projection substitutes $V_\delta = V_{\alpha_0}$. Differentiation in a retained colour different from α_0 gives ordinary derivative naturality; differentiation in V_{α_0} produces

$$(\Phi_{\alpha_0}^{\mathrm{out}})_* \rho_* = \rho_* \big((\Phi_{\alpha_0}^{\mathrm{in}})_* + (\Phi_\delta^{\mathrm{in}})_* \big).$$

The contraction itself is $\rho_* \Phi_{\delta *}$. Its extra term vanishes because $\Phi_{\delta *}^2 = 0$. More generally every colour derivative squares to zero on homology: expand its square as a sum of individual squares and mixed anticommutators, use $\Phi_j^2 = 0$, and differentiate $\mathcal{D}^2 = 0$ in two different variables to see that each mixed anticommutator is null-homotopic. This verifies the substitution rule required when contracting several dummy components in succession.

5.5 Collapsing the four-mark unknot block

Proposition 9 (Four-mark collapse; new combinatorial)**.**

Proposed name: `fourMarkBlockCollapse`.

Suppose an auxiliary grid has a doubled singleton in square (c, r), and has, elsewhere, a persistent component with a row whose O and X are distinct. Expand the singleton by inserting a column immediately to its right and a row immediately above it, and replace it by the four-mark

block

	c	$c+1$
$r+1$	X	O_i
r	O_j	X.

Here j is the old O label and i is the newly introduced label. There is an explicit map from this expanded grid to the original auxiliary grid of degree zero and shifts $(0,0)$, preserving the canonical class. It intertwines the component-colour derivatives when i and j are assigned the original singleton's colour.

Proof. This four-mark pattern is not an ordinary three-corner destabilization, and we do not treat it as one. Instead, exchange its two O markings. The explicit O-switch map F has degree -1 and shifts $(0,-1/2)$. Its target has two doubled singleton squares, at $(c,r+1)$ and $(c+1,r)$. The former carries the new label i, and the latter carries the old label j, because labels travel with their original O rows under this exchange.

Apply Lemma 8 to contract the singleton i, using the persistent component as root. Call that degree-$+1$ map $\mathcal{E}_i$. The composite $\mathcal{E}_iF$ has degree zero and shifts $(0,0)$. The O-switch pairing identity gives

$$(\Phi_i)_*F_*\,\mathrm{can} = \mathrm{can}$$

in the intermediate auxiliary complex. Since $(\mathcal{E}_i)_* = (\rho_i)_*(\Phi_i)_*$ and ρ_i preserves the canonical class, the composite preserves it as well.

Deleting the singleton i means deleting column c and row $r+1$. The retained singleton at $(c+1,r)$ therefore becomes (c,r). Every other inserted-row or inserted-column shift is undone by these deletions. The exact target marking permutations are consequently those of the original auxiliary grid.

Here is also the colour-naturality check; it is needed because the source has two O markings of the singleton's colour. Write A for the expanded source and B for the intermediate two-singleton pair. On homology put

$$K_i = \Phi_i^B F + F\Phi_i^A, \qquad K_j = \Phi_j^B F + F\Phi_j^A.$$

The O-switch calculation gives $K_i = K_j$ on homology, and K_i intertwines each derivative. Using commutation of derivatives on homology and their square-zero identities, we obtain

$$\begin{aligned}\Phi_j^B\Phi_i^B F + \Phi_i^B F(\Phi_i^A + \Phi_j^A) &= \Phi_i^B K_j + \Phi_i^B F\Phi_i^A \\ &= \Phi_i^B K_i + K_i\Phi_i^A = 0.\end{aligned}$$

After applying ρ_i, this is precisely naturality between the surviving Φ_j and the source colour derivative $\Phi_i + \Phi_j$. Other markings of that colour commute through the construction. Derivatives of all other retained colours are natural by the preceding substitution calculation; for the persistent root, its possible extra term vanishes by $\Phi_i^2 = 0$. □

6 The explicit polynomial stabilization maps

Lemma 10 (Polynomial top domain maps; new combinatorial)**.**

Proposed name: `nativeTopDomainMaps`.

For the X:SW stabilization and the incoming type-R formula specified in Section 5, the map R has degree zero, shifts $(0,0)$, and canonical-class identity. Its multivariable chain defect is precisely $(U_i+U_0)L$, where i is the new O and 0 the old O in the stabilized X row. The outgoing formula S below is a polynomial chain map of degree zero and shifts $(0,0)$, preserving the canonical class. Both maps are natural on homology for component-colour derivatives after assigning the new O its

component colour. Together with the supplied commutation maps and corner reduction this proves that naturality for every ordinary move in both directions.

Proof. The map used in the auxiliary contraction is the explicit type-R count, not an arbitrarily chosen projection supplied by the grid-move theorem. We verify its grading and its canonical-class action directly. Write G for the old grid, A for its displayed X:SW stabilization, i for the new O, and 0 for the old O in the row of the stabilized X. The definition and the domain cancellation give, before specialization,

$$D_G R + R D_A = (U_i + U_0)L. \tag{14}$$

Here L counts the trivial domain, together with the nontrivial positive domains with southwest multiplicity $k-1$, the other three multiplicities k, and $2k+1$ moving state coordinates. Its monomial, like that of R, omits U_i. In particular R is a chain map when only $U_i = U_0$ is imposed; all other variables remain independent.

We use the following elementary facts about the domains in this formula. A domain of type R with local multiplicity k has $2k+2$ moving coordinates and decomposes into $2k+1$ empty rectangles. One obtains this decomposition by starting at the initial state's point on the vertical circle through the distinguished intersection p, following the first horizontal boundary segment, and removing the rectangle against that circle. Repeat with the new initial point on that same circle. The local zero-multiplicity condition forces each rectangle to be empty and determines it uniquely. Two successive removals reduce the three equal multiplicities at p by one and remove two moving coordinates. The last step is a rectangle, the case $k = 0$. This is an induction on k. The condition of exactly $2k+2$ moving coordinates excludes an additional boundary component: following the component through p has already used all those coordinates. The same argument with odd complexity proves the corresponding statement for L domains. It also proves finiteness: $2k+2$ is at most the grid size, each boundary segment has finitely many endpoints and two possible circular arcs, and its local multiplicity at p determines the remaining additive constant of the domain.

6.1 The domain cancellation with separate variables

Here are the cuts proving (14). They are included because retaining the old variables is essential. Lift a rectangle of G to the corresponding rectangle of A. Such a rectangle either avoids p, or has p as its sole interior state point; in the latter case its coefficient still omits the new O. Compare an R count followed by one of these lifted rectangles with an empty rectangle of A followed by an R count. A *moving corner* is a point in the symmetric difference of the two states at a step. Let $m = 2k+2$ be the complexity of the R domain.

For clarity the complete cutting list is the following. In this list A_j means that the rectangle avoids p and shares j moving corners with the R domain; B means that p is a corner of the rectangle; C that p lies on an open rectangle edge; and D_j that p is the rectangle's sole interior state point and there are j shared moving corners. A subscript m records the R complexity. The symbol C^- denotes the rectangle on the left of p, and C^+ the rectangle above p. For C^- let δ be the innermost horizontal width of the R domain; for C^+ use its innermost vertical height.

first presentation	other presentation	operation
A_0	A_0	reverse the order of the pieces
A_1	A_1	the other cut at the concave corner
$(A_2)_m$	$(D_0)_{m-2}$	transfer two consecutive turns, $m \geq 4$
B_4 $(m=2)$	the other thin annulus	new-column / old-root-row annuli
$(C^-)_m, \delta = 1$	$(D_1)_m$	add the new-column annulus
$(C^-)_m, \delta > 1$	$(A_3)_{m+2}$	add the new-column annulus
$(C^+)_m, \delta = 1$	$(D_1)_m$	add the old-root-row annulus
$(C^+)_m, \delta > 1$	$(A_3)_{m+2}$	add the old-root-row annulus

The last two rows also have the L-domain presentation described below; the B_4 row has the trivial L-domain presentation.

We explain both exhaustiveness and the indicated constructions. Away from p, three zero quadrants at every state point permit zero, one, or two shared moving corners. Zero corners allow the two pieces to be removed in either order. One shared corner produces a unique concave corner, with the two cuts indicated in the table. With two shared corners, following the boundary removes two successive turns of the R domain and puts p into the lifted rectangle; reversing that cut restores those two turns. This is exactly the A_2/D_0 pair. Three shared corners force the fourth side to go once around a row or column circle: the rectangle has width or height one and the associated innermost R width or height is one. These are the A_3 entries, recovered uniquely by deleting the specified annulus.

If the extra rectangle has p as a corner, its two intermediate coordinates are the unique state points on the two circles through p. The R zero-quadrant conditions force $m = 2$, with the same two moving coordinates. The composite is consequently a thin annulus, either the new-column annulus or the row annulus containing O_0. If p is instead on an open edge, the same quadrant conditions allow precisely a rectangle to the left or above p; the bottom and right alternatives would leave an initial state point in a positive quadrant of the R domain. These are C^-, C^+. Adding the annulus shown in the table completes the boundary across p. When $\delta = 1$, the completed boundary cuts off a rectangle with p as its sole interior state point. When $\delta > 1$, it cuts off a thin rectangle avoiding p and introduces two further turns in the R boundary. The cut starts at the initial state's point in the inserted row or column, so it is unique; deleting that same annulus is the inverse construction. A lifted rectangle having p inside can share only zero or one moving corner, giving exactly D_0, D_1. Finally an empty input rectangle with p strictly inside cannot precede an R domain: follow the R boundary from its initial point on the vertical circle through p to the first exit from that rectangle. Either this initial point is already inside the rectangle, violating emptiness, or that first exit puts one of its corner state points in a positive quadrant of the R domain, violating the three-zero rule. Thus the table accounts for every contribution.

For the first three rows the complete two-chain is unchanged by the cut, and so is every old marking multiplicity. The same holds for the C^- rows except for adding the new-column annulus. That annulus contains only O_i, whose multiplicity in the R factor is omitted from the coefficient. Hence these contributions cancel in pairs with their full old-variable monomials intact.

For B_4, the two annuli have coefficients U_i and U_0. They give $(U_i + U_0)$ times the trivial L-domain coefficient. In a C^+ row the underlying composite itself has three equal multiplicities $k + 1$ at p and southwest multiplicity k; it is an L domain of complexity $m + 1$. Let w be its monomial omitting U_i. The rectangle in the C^+ presentation contains O_i, so its contribution is $U_i w$. Adding the old-root-row annulus gives the other presentation, with contribution $U_0 w$. Their sum is exactly $(U_i + U_0)w$. Every nontrivial L domain is obtained once in this way: cut at its innermost horizontal boundary segment through the distinguished circle, leaving the even-complexity R domain. This is the inverse of the C^+ construction. Together with the trivial case it accounts for the whole right side of (14).

No argument in this cancellation used X markings. It needs only one O per row and column, the new O in the northeast square, and the designation of the old O in the row immediately below it. It therefore also holds when that old O happens to occupy the southwest square. This observation justifies the temporary O-only normal form used below, as well as auxiliary coincident markings elsewhere. Substituting $U_i = U_0$ proves the claimed multivariable chain equation without identifying any further variables.

6.2 Gradings and polynomial coefficients

Cyclically choose coordinates so that G has size n, the stabilized X lies in its last column and row 0, and the new column and new row of A are column n and row 1. Thus $p = (n, 1)$. If y contains

p and $e(y)$ deletes that coordinate, direct substitution in J gives

$$M_A(y) = M_G(e(y)) - 1, \qquad A_A(y) = A_G(e(y)) - 1.$$

For the first equality, the extra contributions to $J(y,y), -2J(y,O_A), J(O_A,O_A)$ are respectively $1, -3, 1$. The analogous change of M_X is zero, proving the second equality after the change of grid size is included. These calculations use only the row and column permutations. They remain true if an old O coincides with the chosen old X.

Let $q : x \to y$ be a type-R domain with three local multiplicities k. Its multiplicity at the new O is k, and that at the southeast X is $k+1$. Put $w(q) = \sum_{j \neq i} n_{O_j}(q)$. Summing the rectangle grading formulas over the decomposition above gives

$$M_A(x) - M_A(y) = 2k + 1 - 2(k + w(q)) = 1 - 2w(q).$$

Consequently $T^{w(q)}e(y)$ has exactly the degree of x. For the second filtration coordinate the same calculation gives

$$A_G(e(y)) - w(q) - A_A(x) = 1 + k - n_X(q) = -\sum_{X \neq X_{\mathrm{SE}}} n_X(q) \leq 0.$$

All exponents $w(q)$ are nonnegative. Thus R has degree zero and shifts $(0,0)$, also for auxiliary marking pairs with other coincident O and X markings.

6.3 Compatibility with O switches away from the new column

The following argument concerns only O diagrams, so a temporary coincidence with an X marking has no effect. Exchange O markings in two adjacent old columns. They are still adjacent in A, since its new column is last. Let F_G, F_A be the respective diagonal O-switch maps, and let R, R' be the type-R counts before and after this change. Their domain sets are identical: the definition of a type-R domain does not refer to O markings.

Write $\varepsilon_A, \varepsilon_G$ for the switch indicators on the two grids, reserving $e(y)$ for deletion of p. For any such domain $q : x \to y$, the same boundary computation as for a rectangle gives

$$n_{O'}(q) - n_O(q) = \varepsilon_A(y) - \varepsilon_A(x).$$

It can equivalently be obtained by summing the rectangle identities over its decomposition. The new O has not moved. Since the switch uses old columns, $\varepsilon_A(y) = \varepsilon_G(e(y))$. Comparing the monomial of each individual domain therefore gives the exact identity

$$R'F_A = F_G R. \tag{15}$$

The same assertion holds for the reverse diagonal switch map.

Choose, among the two exchanged labels, a label $j \neq 0$. There is always such a label. Differentiation of (14) after $U_i = U_0$ proves that R and R' intertwine Φ_j on homology. Using (15) in the definition of the degree-zero Clifford maps K_j consequently gives

$$R'_*(K_j^A)_* = (K_j^G)_* R_*. \tag{16}$$

This remains valid when label 0 itself is one of the exchanged markings: the detecting label j is the other one, so its variable remains independent of $U_i = U_0$.

6.4 An explicit normal-form evaluation

Bubble-sort the old O permutation to $O_G(c) = n - 1 - c$, using adjacent old columns only. Each K_j in (16) preserves the canonical class and is invertible on homology. It follows that the desired nonvanishing for R is equivalent to nonvanishing in this one normal form. The old O in row 0 is then in the last column; hence the chosen old X would coincide with it. This temporary O-only calculation is legitimate: the type-R domain identity and (16) use no X markings, and canonical classes are defined from the O complex using any disjoint X permutation.

The stabilized O permutation in this normal form is

$$O_A = (n, n-1, \dots, 2, 0, 1).$$

It is obtained from the size-$(n+1)$ antidiagonal by interchanging its last two O markings. Write

$$\theta = (0, n, n-1, \dots, 2, 1), \qquad \theta_G = (0, n-1, n-2, \dots, 1).$$

The antidiagonal canonical cycle is θ. For this last O switch the degree-minus-one diagonal map is $F(x) = T^{1-\mathbf{1}_{\{x(n)=1\}}}x$, so it sends θ to itself. Every derivative of θ in the antidiagonal complex is zero already on chains, since it would have positive degree and polynomial coefficients, while that complex has no positive-degree polynomial generators. The Clifford identity therefore identifies the canonical cycle of A as

$$z = \Phi_i^A \theta.$$

Here this cycle can be written completely explicitly. For $0 \le b \le n-1$, let x_b be obtained from θ by interchanging its coordinates in columns b, n. The empty rectangles from θ through the new O are exactly the n rectangles with southwest corner $(n, 1)$ and northeast corner $(b, \theta(b))$, using the circular intervals from n to b and from 1 to $\theta(b)$. Indeed, a rectangle through that O which did not start on column n would contain the state point $(n, 1)$ in its interior. For the displayed rectangles all intervening antidiagonal state points and all old O markings lie outside. Each contains the new O once and no other O. Hence

$$z = \sum_{b=0}^{n-1} x_b.$$

Every x_b has Maslov degree zero. The only possible degree-zero polynomial output of R in the old antidiagonal complex is θ_G with coefficient 1, since its other states have strictly negative Maslov degree. The corresponding stabilized terminal state is precisely θ.

Since x_b differs from θ in two coordinates, a type-R domain from x_b to θ must have $k = 0$; it is a rectangle with northwest corner $p = (n, 1)$. For $b = 0$ the unique such rectangle is the single square in the new column and row 0. It contains no O marking and contributes θ_G. For every $b \ge 1$, the only rectangle with the required southeast multiplicity at p uses the circular intervals from column n to column b and from row $\theta(b)$ to row 1. It contains the unchanged state point $(0, 0)$ strictly in its interior, so it is not a type-R domain. Therefore

$$Rz = \theta_G.$$

This proves nonvanishing in the normal form. Transport through (16) proves it for every O permutation. Since R has degree zero and preserves the polynomial subcomplex, Lemma N now yields

$$R_* \operatorname{can}(A) = \operatorname{can}(G). \tag{17}$$

This establishes the property of this specific polynomial formula without using any unspecified stabilization map or any comparison with holomorphic Floer homology.

6.5 The outgoing top map and colour naturality in both directions

For completeness, the stabilization direction has an equally explicit polynomial representative. Given a state x of G, extend it by p to a state $\widetilde{x}$ of A. Define

$$Sx = \sum_{y} \sum_{q:\widetilde{x}\to y} \prod_{j\neq i} U_j^{n_{O_j}(q)} y,$$

where the inner sum is over nonnegative outgoing domains satisfying the same three-zero rule away from p, having multiplicity $k+1$ in the northeast square at p and k in the other three squares, and having $2k+2$ moving coordinates. These domains have the same finite boundary description and rectangle decomposition as the incoming domains, with arrows reversed and the local vertical coordinate reflected. This description is a definition of the formula, not an instruction to reflect the grid or its complex.

The boundary cuts in the preceding table apply with their arrows reversed and with north and south interchanged. The two annuli in the exceptional rows are now the new-column and new-row annuli. Each contains only the new O, and in each presentation its multiplicity belongs to the outgoing domain, whose monomial omits that O. Their coefficients are therefore equal. More explicitly, the reflected C^+ configuration has an extra rectangle below p. That rectangle contains no new O. Its union with the outgoing domain has northeast, southeast, and southwest multiplicities $k+1$ and northwest multiplicity k. Adding the new-row annulus changes the northeast multiplicity to $k+2$ and the other three to $k+1$. These are the two alternative presentations; both have exactly the same old-variable monomial. The complexity-two exception has the same two equal coefficients. The other rows cancel by equal total domains or by the new-column annulus, exactly as before. Consequently

$$D_A S = S D_G \tag{18}$$

already with *all* O variables independent, interpreting the source as extended by the unused variable U_i.

For an outgoing domain with parameter k, its multiplicity at O_i is $k+1$ and at the southeast X is k. Its $2k+1$-rectangle decomposition and the displayed state-extension grading formulas give

$$M_A(y) - 2\sum_{j\neq i} n_{O_j}(q) = M_G(x), \qquad A_A(y) - \sum_{j\neq i} n_{O_j}(q) - A_G(x) = k - n_X(q) \leq 0.$$

Thus S has degree zero and shifts $(0,0)$.

Its canonical-class action follows by the same normal-form method, and can be checked particularly simply. The O-switch boundary identity gives

$$F_A S = S' F_G$$

term by term. Differentiating (18) and using this exact identity gives the corresponding K-naturality on homology. Therefore sort the old O permutation as above and evaluate S on θ_G. Its extension is θ.

An outgoing domain starting at θ cannot have $k > 0$. Indeed, the southeast unit square at $p = (n, 1)$ is the northwest square at the initial state point $(0, 0)$, using the circular last column. Its multiplicity is k. The three-zero rule at $(0, 0)$ would then leave a positive northwest quadrant alone. This has the boundary sign of a terminal corner, whereas $(0, 0)$ is an initial state point; if that point were also terminal, all four quadrants would have to vanish. Both possibilities contradict $k > 0$. Hence $k = 0$. The domains are precisely the n rectangles from θ to x_b already enumerated above. Their only O is the new one, whose factor is omitted. We conclude

$$S\theta_G = \sum_{b=0}^{n-1} x_b = z.$$

Transport by K proves that this specific map S preserves the canonical class for every old O diagram.

Finally identify the new O variable with the variable of the component on which the stabilization is performed, and give every other component its own variable. Equations (14) and (18) are now chain equations over that colour ring in both directions. Differentiating in any colour variable proves naturality for the sum of its marking derivatives. For a commutation the empty-pentagon formula retains the complete O monomial of each polygon; the alternate rectangle–pentagon cut has that same monomial, so its chain equation is valid over the colour ring too. Cyclic permutations simply relabel these monomials. The granted corner reduction expresses every other stabilization orientation using these corner maps and such commutations and permutations. Every extra O in that reduction is on the same physical component as the O with which its variable is identified. Composing the differentiated identities therefore proves colour naturality for every ordinary move used in the construction, in both directions, on arbitrary homology classes.

□

7 Explicit relations for signed word grids

Lemma 11 (Signed word-grid relations; new combinatorial)**.**

Proposed name: `signedWordGridRelations`.

Every finite derivation by Artin relations, far commutations, free inverse cancellation or insertion, and cyclic whole-word rotations has a finite ordinary-grid realization under the signed gadget convention above. All local replacements work in arbitrary signed contexts with the indicated port incidences. A rank becoming unused is retained as the specified four-mark block. For an Artin/free derivation whose initial and final words contain every generator, the external guard in the proof makes every word in the initial normalization contain every generator. All displayed rules specify legal noninterleaving exchanges and genuine three-corner stabilization patterns.

Proof. The negative-letter gadget has outgoing heights $(r+\frac{1}{2}, r+1, r)$ and incoming heights $(r+1, r, r+\frac{1}{2})$ in its three consecutive columns. Integer-height arcs join consecutive occurrences cyclically, exactly as for positive letters. The following constructions work in an arbitrary signed-word context, and use no topological equivalence theorem.

7.1 Two elementary grid rules

We distinguish strict noninterleaving commutations from the case of a shared endpoint. The latter case can also be performed by ordinary grid moves, as follows. Suppose consecutive columns A, B have

$$(O_A, X_A) = (r, p), \qquad (O_B, X_B) = (q, r).$$

First assume $p \neq q$. The row r joins the consecutive columns A, B. Move this row toward row p, along the arc of the row circle from r to p which avoids q, stopping adjacent to p. Every row actually crossed has both its markings outside A, B, so each exchange is a strict noninterleaving row commutation. The block with columns A, B and rows r, p has precisely the three markings

$$O(A, r), \quad X(A, p), \quad X(B, r).$$

Destabilize, deleting column A and row r, and replacing the two X's by $X(B, p)$. Now stabilize this X using a new column on the opposite side of B: restore row r and put

$$(O_B, X_B) = (q, r), \qquad (O_A, X_A) = (r, p)$$

in the order B, A. Move the short row r back through the same rows in reverse order. This is exactly the required exchange of A, B.

If $p = q$, the two columns form a two-row unknot block. Insert a row t next to p and a column D immediately before A, by the X stabilization

$$D : (O = t, X = p), \quad A : (O = r, X = t), \quad B : (O = p, X = r).$$

Exchange A, B by the preceding rule, then exchange D, B by its O/X-interchanged version. Both exchanges have distinct nonshared endpoints. The order is now B, D, A. Destabilize the block $O(D, t), X(D, p), X(A, t)$, deleting D, t. The result is B, A with exactly the required markings. All other possible shared-endpoint patterns follow by interchanging O, X and/or reversing the names of the two columns. Transposing this rule proves its row version. Henceforth "exchange" means a strict commutation or this fully specified replacement when endpoints coincide.

The second rule contracts a bivalent column. Suppose column C carries O in row u and X in a different row v, with u, v adjacent, and let E be the other column carrying an X in row u. Slide C around the column circle until immediately before E. Its marking interval has consecutive endpoints, so all intervening exchanges are noninterleaving or shared-endpoint exchanges just treated. If $O(E) \neq v$, destabilize the three-corner block

$$O(C, u), \qquad X(C, v), \qquad X(E, u),$$

deleting C, u and replacing $X(E, u)$ by $X(E, v)$. This joins the two horizontal arcs. If $O(E) = v$, the block is instead an isolated 2×2 unknot and is retained as such.

The inverse rule subdivides any horizontal arc: at its X endpoint insert an adjacent row and column using the reverse displayed three-corner pattern, then slide the resulting short vertical column to the desired position. Thus an incoming and an outgoing port which were the same horizontal arc can always be separated by an ordinary stabilization, with a bivalent column outside a specified local word block. After a local construction this subdivision can be contracted unless that entire strand has become unused, in which case its 2×2 unknot block is retained.

7.2 Choice of row orders at a repeated height

At an integer height the horizontal arcs run between successive occurrences of that rank. Their interiors in the column circle are disjoint. At a half-integer height the arcs lie in distinct three-column letter blocks and again have disjoint interiors. Therefore any two rows of the same height can be exchanged after bringing them next to one another: their endpoint pairs are nonalternating, with shared endpoints covered by the first rule above. Bubble-sort consequently realizes every desired tie-break by ordinary grid moves.

7.3 Cancellation of consecutive inverse letters

Consider $\sigma_r \sigma_r^{-1}$ and name its six consecutive columns $A_0, A_1, A_2, B_0, B_1, B_2$. Denote its horizontal rows by

height	row	arc
r	a	previous occurrence $\to A_0$
r	b	$A_1 \to B_1$
r	c	$B_2 \to$ next occurrence
$r + \frac{1}{2}$	p	$A_0 \to A_2$
$r + \frac{1}{2}$	q	$B_0 \to B_2$
$r + 1$	d	previous occurrence $\to A_1$
$r + 1$	e	$A_2 \to B_0$
$r + 1$	f	$B_1 \to$ next occurrence.

Thus the marking lists in these six columns are

$$O = (p, b, e, q, f, c), \qquad X = (a, d, p, e, b, q).$$

If $a = c$ or $d = f$, subdivide that horizontal arc outside this six-column block by the second elementary rule. These rows are then distinct. The subdivision is remembered as belonging to a rank which will become unused after the cancellation.

Choose p, q to be the last two rows at height $r + \frac{1}{2}$, in that order, and e to be the first row at height $r + 1$. Perform:

1. The rows q, e and columns A_2, B_0 contain $O(A_2, e), X(B_0, e), O(B_0, q)$. Delete B_0, e by destabilization, replacing these by $O(A_2, q)$.
2. Exchange A_2, B_1. The former has endpoints p, q, while the latter has endpoints b, f, so the intervals are nested. Now A_1, B_1 are adjacent and row b is a short row. Move it next to d. Destabilize $O(A_1, b), X(A_1, d), X(B_1, b)$, deleting A_1, b and putting $X(B_1, d)$.
3. Destabilize the block $O(A_2, q), X(A_2, p), X(B_2, q)$, deleting A_2, q and putting $X(B_2, p)$. The rows p, q are still adjacent.
4. Exchange B_1, B_2. Both endpoints of B_1 are in the upper integer-height group, and the interval of B_2 runs from the lower group to p, so they are disjoint. Now row p has the adjacent endpoints A_0, B_2. Move it next to c and destabilize $O(A_0, p), X(B_2, p), O(B_2, c)$, deleting B_2, p and putting $O(A_0, c)$.
5. The remaining two local columns are $A_0 : (O = c, X = a)$ and $B_1 : (O = f, X = d)$. Bring a, c together within their common height group and apply bivalent contraction to A_0; do the same with d, f and B_1.

For a rank which still occurs outside the cancelled pair, the final contraction is a three-corner destabilization. For an unused rank, the remembered subdivision column and this final column form precisely an isolated 2×2 unknot; retain that block as the resolution of the unused strand. There is no death or saddle in this case. The two columns of this block can be brought together, and the block can be moved to any prescribed auxiliary position using the elementary exchanges above.

When both ranks persist, the construction uses exactly six destabilizations, with exchanges and harmless tie-break changes. All six gadget columns have disappeared, and the two surviving integer-height arcs are exactly the arcs formed by skipping the two cancelled occurrences. Columns outside the block retain their cyclic order; if necessary make one cyclic column permutation to restore the chosen cut. The resulting grid is the signed word grid with the pair removed, up to tie-break changes. This checks the endpoint by marking incidence, not by a link-isotopy assertion.

For $\sigma_r^{-1}\sigma_r$, reverse the local row coordinate, applying the same rule with upper and lower ranks interchanged. This means conjugating each explicit move by row reflection, not performing a reflection as a grid move. Commutations and the eight stabilization patterns are closed under that reflection, so the reflected sequence is again a sequence of the permitted moves. Inverse-pair insertion is the reverse of the corresponding sequence.

7.4 The three-letter Artin relation

For the source block $\sigma_r\sigma_{r+1}\sigma_r$ label its columns $0, 1, \ldots, 8$. Each outgoing row is labelled by its starting column. Let $9, 10, 11$ denote the incoming port rows at integer heights $r, r + 1, r + 2$. If any such port is the same arc as its outgoing port, first subdivide that arc outside the block as above, so all twelve distinguished rows are distinct. No other subdivision is required.

The initial X incidences in the nine columns are

$$X = (9, 10, 0, 2, 11, 3, 1, 4, 6), \qquad O(c) = c.$$

Choose tie-breaks so that the distinguished rows have the relative order

$$(1, 7, 9,\ 0, 6,\ 2, 4, 8, 10,\ 3,\ 5, 11).$$

The groups displayed correspond respectively to heights $r, r+\frac{1}{2}, r+1, r+\frac{3}{2}, r+2$. Outgoing port rows $7, 8, 5$ end at columns outside the block; incoming port rows $9, 10, 11$ start at columns outside the block.

In the following table $C_{a,b}$ exchanges the columns with persistent labels a, b, and $R_{a,b}$ exchanges the rows with those labels. Each column pair is adjacent at its indicated step. For rows, first slide the internal row across any nondistinguished rows between the indicated pair, then perform the exchange. Such intermediate rows have both endpoints outside the local column block, so these preliminary slides are strict noninterleaving commutations.

The last two columns verify noninterleaving at every step. For a column exchange the numbers are positions in the current order of the twelve distinguished rows. For a row exchange they are positions in the current order of the nine local columns; $*$ is any outside column, all of which lie in the complementary interval after position 8 when the circle is cut immediately before the local block. The endpoint pairs are disjoint, nested, or share an endpoint. Shared endpoints use the fully specified elementary rule above.

1	$C_{0,1}$	$(2,3)$	$(0,8)$
2	$R_{1,7}$	$(0,6)$	$(7,*)$
3	$R_{6,2}$	$(6,8)$	$(2,3)$
4	$C_{5,6}$	$(9,10)$	$(1,5)$
5	$C_{5,7}$	$(9,10)$	$(0,6)$
6	$C_{5,8}$	$(9,10)$	$(5,7)$
7	$C_{2,3}$	$(3,4)$	$(4,9)$
8	$C_{2,4}$	$(3,4)$	$(6,11)$
9	$C_{2,6}$	$(3,4)$	$(1,5)$
10	$C_{2,7}$	$(3,4)$	$(0,6)$
11	$C_{4,6}$	$(6,11)$	$(1,5)$
12	$R_{6,4}$	$(3,7)$	$(4,5)$
13	$R_{2,4}$	$(2,6)$	$(4,5)$
14	$R_{6,8}$	$(3,7)$	$(7,*)$
15	$R_{6,10}$	$(3,7)$	$(0,*)$
16	$R_{6,3}$	$(3,7)$	$(2,8)$
17	$C_{3,6}$	$(5,8)$	$(1,9)$
18	$C_{0,6}$	$(2,3)$	$(1,9)$
19	$R_{1,9}$	$(0,1)$	$(2,*)$
20	$R_{1,0}$	$(0,1)$	$(2,6)$
21	$C_{3,4}$	$(5,8)$	$(4,11)$
22	$C_{0,4}$	$(1,2)$	$(4,11)$
23	$R_{1,4}$	$(0,1)$	$(2,5)$
24	$C_{6,4}$	$(4,9)$	$(3,11)$
25	$C_{3,7}$	$(5,8)$	$(0,3)$
26	$R_{1,2}$	$(0,2)$	$(5,6)$
27	$C_{3,2}$	$(4,8)$	$(2,4)$
28	$R_{1,8}$	$(0,2)$	$(7,*)$
29	$R_{1,10}$	$(0,2)$	$(0,*)$

Every row exchange includes an internal row: these are rows $0, 1, 2, 3, 4, 6$, whose two endpoints always remain among the nine local columns. Thus the preliminary slides past entirely external rows are always permitted. In rows of the table involving a port, the displayed noninterleaving condition holds for every possible outside endpoint, not just for one choice of closure.

After step 29 the column order is

$$(1, 4, 6, 0, 7, 2, 3, 8, 5)$$

and the distinguished row order is

$$(7, 9,\ 0,\ 4, 2, 8, 10,\ 1, 3,\ 6, 5, 11).$$

The five groups now have heights $r, r+\frac{1}{2}, r+1, r+\frac{3}{2}, r+2$. Reinsert the entirely external rows in height order: only internal rows need to pass them, and those exchanges are harmless as just proved. The three incoming and three outgoing port rows never change their heights or their order relative to external rows.

To check the endpoint, identify the new three-letter columns with the old labels by

$$(\text{new } 0, \dots, 8) \longmapsto (1, 4, 6, 0, 7, 2, 3, 8, 5).$$

The unchanged O/X incidence, with the final row-height groups just listed, is precisely the gadget sequence $\sigma_{r+1}\sigma_r\sigma_{r+1}$. In particular the three incoming ports and the three outgoing ports are the same, so this is a replacement in an arbitrary surrounding word. Contract any temporary port subdivisions, then choose the specified final tie-break. This finishes a completely local grid proof of the Artin relation.

The all-negative relation follows by reflecting the local height coordinate and conjugating each listed grid move by that reflection. Alternatively it follows by inserting inverse pairs, applying the positive relation, and cancelling inverse pairs. No mixed-sign relation beyond these generators of the group congruence is needed.

7.5 Far commutation and cyclic rotation

If $|i-j| \geq 2$, every vertical marking interval in the three columns of a $\sigma_i^{\pm 1}$ gadget lies in the height strip $[i, i+1]$, and those of a $\sigma_j^{\pm 1}$ gadget lie in $[j, j+1]$. These strips are disjoint. Move the three columns of the second block past the three columns of the first block, one at a time, for nine strict column commutations. The rank sets are disjoint, so the integer-arc incidence does not change except for the indicated block order. This proves far commutation for every choice of signs.

For a cyclic word rotation $uv \mapsto vu$, cyclically permute the columns by $3\ell(u)$. Integer-arc incidence is cyclic and hence unchanged, and the row-height order is unchanged. Correcting the tie-break by row exchanges gives the chosen word grid. Any auxiliary unused-strand blocks can be kept at their prescribed positions by sliding their short rows and columns as above.

Consequently every finite derivation using free inverse cancellation, Artin relations, far commutation, and cyclic word rotations has an explicit finite realization by ordinary grid moves. Each generator of the derivation is replaced by the rule specified here; there is no appeal to the topological statement that isotopic link grids are connected by moves.

7.6 Moving a consecutive active block to the first ranks

Set $\rho = \sigma_0\sigma_1\cdots\sigma_{S-2}$. For $0 \leq i \leq S-3$ the following calculation uses only the two positive Artin relations just realized:

$$\begin{aligned}\rho\sigma_i &= (\sigma_0\cdots\sigma_{i-1})\sigma_i\sigma_{i+1}\sigma_i(\sigma_{i+2}\cdots\sigma_{S-2}) \\ &= (\sigma_0\cdots\sigma_{i-1})\sigma_{i+1}\sigma_i\sigma_{i+1}(\sigma_{i+2}\cdots\sigma_{S-2}) \\ &= \sigma_{i+1}\rho.\end{aligned}$$

The first equality moves the final σ_i left past indices at least $i+2$; the last moves σ_{i+1} left past indices at most $i-1$. Empty strings at the ends are omitted. Appending ρ^{-1} and freely cancelling proves $\rho\sigma_i\rho^{-1} = \sigma_{i+1}$ by a finite word derivation. Taking inverses gives the same identity for σ_i^{-1}. Consequently, if β uses only indices $0, \dots, a-2$ and $a+k \leq S$, iteration gives

$$\mathrm{shift}_k(\beta) = \rho^k\beta\rho^{-k}.$$

This identity is obtained letter by letter, inserting inverse pairs between consecutive letters when needed. It therefore has an explicit ordinary-grid realization by the preceding rules. In particular it applies to a one-crossing factor, as well as to a local positive word on any consecutive block of active strands.

7.7 A guard preserving the ambient number of strands

When a finite word derivation is to be realized before the scissor construction, its intermediate grids can all be kept on the prescribed S ranks, with every generator present, as follows. Put $p = \rho\rho^{-1}$. This literal word contains every generator and hence uses every rank, and represents the identity. Suppose α contains every generator and a finite derivation by local Artin and free relations changes α into a product $Q = q_1 \cdots q_r$, where every q_j has been padded as $p\widetilde{q_j}p$. Perform the derivation in the sequence

$$\alpha \longrightarrow p\alpha \longrightarrow pQ \longrightarrow Q.$$

The first arrow inserts inverse pairs in front of α; its original letters keep every generator present during that insertion. In the middle arrow, keep the displayed external copy of p unchanged and perform the chosen derivation entirely in the remaining suffix. The local grid rules allow arbitrary surrounding words, so this is exactly the prescribed realization with a fixed guard. It is not necessary to rotate this external guard through any factor. For the last arrow cancel its central inverse pairs successively. The individually padded suffix Q keeps every generator present until that external copy has disappeared.

If a preceding comparison is only up to whole-word conjugacy, first absorb that conjugacy into every factor conjugator: $c(q_1 \cdots q_r)c^{-1} = \prod_j (cq_jc^{-1})$. Inserting $c^{-1}c$ between factors is a finite free derivation. This gives exactly the Artin/free suffix derivation used here; no cyclic rotation of a proper substring is being asserted to be a word relation. The conjugator can itself be read from any finite comparison that uses whole-word rotations. Maintain an identity $\alpha = cvc^{-1}$, starting with c empty and $v = \alpha$. A local Artin or free step changes v and leaves c fixed. For a rotation $v = ab \mapsto ba$, replace c by ca:

$$c(ab)c^{-1} = (ca)(ba)(ca)^{-1},$$

where the equality is realized by inserting aa^{-1} immediately after the displayed b. This records a finite derivation using only local Artin and free relations. At the end absorb the recorded c into the factor conjugators as above, before inserting the guard.

Thus the guard is absent before any scissor, factor exposure, or marked conjugacy cleanup. Each remaining factor has the form

$$p\gamma\eta\gamma^{-1}p = (p\gamma)\eta(p\gamma)^{-1},$$

as a literal word, because $p^{-1} = p$. Its padding is consequently removed by the same marked inverse-pair cancellation as its conjugator. No extra cyclic rotation of a guard, or assumption about a marked component during the guarded derivation, is required.

□

8 Cancelling inverse letters while retaining an attached summand

Lemma 12 (Marked inverse cancellation; new combinatorial)**.**

Proposed name: `markedInverseCancellation`.

Let a closed signed word have a marked horizontal integer-rank arc in a nontrivial permutation cycle, with an arbitrary prepared grid H inserted in that arc in the frame specified below. A consecutive inverse pair at the marked cut can be cancelled by the displayed ordinary moves while

preserving all internal marking data of H. The mark changes by the cancelled letter's transposition, and the exact frame is restored. Iteration reduces a marked $\gamma\eta\gamma^{-1}$ to η, with tracked rank $\pi(\gamma)^{-1}(k)$ from initial rank k. Unused unmarked ranks become precisely the stated four-mark components. These moves are natural for component-colour derivatives.

Proof. Here a protected grid H may contain other components in addition to its marked component. Use its supplied seam preparation, so that

$$X_H(0) = 0, \qquad O_H(h-1) = 0.$$

These are exactly the right-hand preparation conditions for the binary seam. No further reordering of the protected grid is required. In particular, its last X need not be in its top row.

8.1 The frame and its elementary exchanges

In a base grid choose a horizontal arc from $O(P, b)$ to $X(Z, b)$. The framed insertion of H into that arc has the following data: the base column Z is retained, with its X moved from b to a new row t; the first column of H has X in row b; its last column has O in row t; and every other marking of H is in an internal H row. Thus the two exterior rows are

$$b : P \longrightarrow H_{\text{first}}, \qquad t : H_{\text{last}} \longrightarrow Z.$$

The row circle is ordered as

$$(t,\ \text{all other base rows in their old cyclic order after } b,\ b,\ H_1, \ldots, H_{h-1}). \tag{19}$$

The H columns form one contiguous block and retain their internal order.

This frame is obtained from an existing seam by explicit ordinary moves. The marked column Z of the base grid has been removed in that seam: $X(Z, b)$ is represented by the first H column's X, and $O(Z, u)$ by the last H column's O. O-stabilize at this last O, inserting a column C immediately to its right and a row t next to u, with

$$O(H_{\text{last}}, t), \quad X(C, t), \quad O(C, u).$$

Regard C as the restored base column Z. Its O is exactly the O that the base column originally had. The row t has consecutive column endpoints H_{last}, C, so it can be moved to immediately after b in the cyclic order of the base rows, using strict exchanges and the explicit shared-endpoint rule. Then gather all H internal rows between b, t, preserving their order. To justify every exchange in this last operation, an H internal row has both endpoints in the H column block. An exterior row has both endpoints outside that block, or one endpoint at its first or last column. These pairs are always disjoint, nested, or share an endpoint. Thus every exchange is one already explicitly realized by ordinary grid moves. This proves (19) from the seam, without an isotopy theorem.

The frame has an equally explicit inverse. In an exact frame move the H block immediately before the restored base column Z, using the column exchanges established below. The output row t is then short, with endpoints H_{last}, Z. Move it next to $u = O(Z)$ and O-destabilize the block

$$O(H_{\text{last}}, t), \quad X(Z, t), \quad O(Z, u),$$

deleting Z, t and putting $O(H_{\text{last}}, u)$. This restores exactly the seam marking formula. Choose the column cut at the removed marked column and the base row cut after the input row, then gather the internal H rows above the base rows. The same permitted row exchanges give the actual binary seam, with its prescribed internal row orders.

Two elementary facts will be used repeatedly.

1. In the exact frame, every H column can be exchanged with every base column. Cut rows at t in (19). All base column endpoints lie between t and b. Every H column except its last has both endpoints between b and H_{h-1}; the last H column joins t to its own internal X row, which is above b, and therefore encloses the whole base interval. The only endpoint equalities are handled by the shared-endpoint rule. Hence the complete H block can be moved to any gap in the base column order, without changing that order or the order within H.

2. Even after the two exterior port rows have changed, every internal H row can be exchanged with every exterior row. The column argument in the preceding paragraph still applies: the only exterior endpoints in the H block are at its first and last columns. Thus the H internal rows can be placed in any desired interval of the row circle, preserving the order of all exterior rows and their own internal order. If the two port rows are adjacent in the base order, putting the H internal rows between the input and output ports restores the exact frame.

The following boundary absorptions also use only three-corner destabilizations. If a base column L immediately precedes H and has $O(L, b), X(L, d)$, while the H input is $X(H_{\text{first}}, b)$, move the short row b next to d and delete L, b. The H input becomes $X(H_{\text{first}}, d)$. Dually, if a base column R immediately follows H and has $X(R, t), O(R, f)$, while $O(H_{\text{last}}, t)$ is the H output, move the short row t next to f and delete R, t. The H output becomes $O(H_{\text{last}}, f)$. The other corner is empty in each case. These are the usual X- and O-destabilization patterns, respectively.

8.2 Marked lower connector of a positive–negative pair

Use the six-column notation of the unmarked cancellation theorem:

$$O = (p, b, e, q, f, c), \qquad X = (a, d, p, e, b, q)$$

in columns $A_0, A_1, A_2, B_0, B_1, B_2$. Insert framed H into the lower connector $b : A_1 \to B_1$. Thus B_1 now has X in the frame's output row t, and the H ports are b, t. Apply the preliminary port subdivision of the unmarked rule if the unmarked lower ports a, c coincide. The marked upper ports d, f are distinct in the application: otherwise the tracked marked component would be the isolated rank which becomes unused, contrary to its nontrivial permutation cycle.

First do the ordinary destabilization deleting B_0, e and putting $O(A_2, q)$, and exchange A_2, B_1. Beforehand H may be moved to a gap outside these columns by the first framing fact. Move its block into the gap between A_1, B_1. Absorb A_1, b into its left boundary, giving H input d, and absorb B_1, t into its right boundary, giving H output f. Thus H now replaces the upper bivalent column which occurs after the corresponding two stages of the unmarked cancellation.

Place all H internal rows just above the upper integer-height rows d, f, using the second framing fact. Consequently every endpoint of an H column is above the lower and middle rows of the remaining A_0, A_2, B_2 portion. Delete A_2, q by the ordinary three-corner destabilization, placing $X(B_2, p)$. Exchange the entire H column block past B_2: its intervals and that of B_2 are disjoint. Row p now has consecutive endpoints A_0, B_2, so perform the ordinary destabilization deleting B_2, p and putting $O(A_0, c)$. Finally contract the unmarked lower bivalent column A_0.

If the unmarked rank persists, all six original letter columns have been removed by six destabilizations. The H columns and internal rows are unchanged in either case. Its new input and output rows are precisely d, f, the two portions of the upper-rank arc obtained by deleting the inverse pair. This is the same insertion on that arc, before the reframing specified below. If the unmarked lower rank becomes unused, the last bivalent contraction instead retains A_0 and its preliminary subdivision column as the 2×2 auxiliary block of the unmarked rule; the other five letter columns have been removed.

8.3 Marked upper connector of a positive–negative pair

Now insert framed H into $e : A_2 \to B_0$, so B_0 has X in row t. Subdivide the unmarked upper arc outside the block if $d = f$. The marked lower ports a, c are distinct by the same nontrivial-cycle argument. Move H between A_2, B_0. Absorb A_2, e at its left boundary, giving H input p, and absorb B_0, t at its right boundary, giving H output q. H now replaces the middle bivalent column occurring after the first stage of the ordinary cancellation.

Place its internal rows immediately above the middle row q and below *every* upper integer-height row. The interval of B_1, whose endpoints are b, f, then contains every H column interval. Exchange the whole H block past B_1. The columns A_1, B_1 are now adjacent, so the ordinary short-row move and destabilization delete A_1, b and put $X(B_1, d)$.

Absorb B_2, q at the right boundary of H, giving output c. Its input is still p. All H column endpoints are now below the upper-rank endpoints d, f of B_1, so exchange B_1 past the entire H block by disjoint-interval commutations. Absorb A_0, p at the left boundary of H, giving input a. Contract the unmarked upper bivalent column B_1.

Again exactly six original letter columns have disappeared if the unmarked rank persists. H is unchanged internally and is now inserted on the lower-rank arc with ports a, c. An unused unmarked upper rank is treated by retaining its ordinary 2×2 auxiliary block, consisting of B_1 and the preliminary subdivision column; the other five letter columns have disappeared.

8.4 Reframing and iteration

The port rows d, f in the first case, or a, c in the second, belong to one integer-height group in the word with the pair removed. They can be made adjacent in either prescribed tie order. Here is the needed noninterleaving check even when H is present. Let P, Q be the preceding and following occurrences of this rank. The remaining H block lies in the cyclic column interval from P to Q which formerly contained the six-letter-gadget columns. The two port intervals are $P \to H_{\text{first}}$ and $H_{\text{last}} \to Q$. All other rows at this integer height are arcs between different consecutive occurrences; their interiors are disjoint from that interval, or contain it after choosing a planar cut. Hence each of the two port rows exchanges with every such tie row by a strict or shared-endpoint exchange.

Put the input port immediately before the output port in the base cyclic order. By the second framing fact move the H internal rows between them. This restores (19) exactly and allows the procedure to be repeated for the next inverse pair.

The negative–positive case is obtained by reversing the local row coordinate in every displayed move, interchanging upper and lower ranks. This conjugates the move formulas, not the protected summand: its internal rows are labelled in the opposite order while describing the reflected formulas, and those labels are then translated back. Every actual step is still an ordinary grid move and the protected marking data are unchanged. A marked rank not touched by the cancelled letters is an exterior port: move H away from the local block by the first framing fact and apply the unmarked cancellation. Thus every case is covered. The rank carrying the mark changes by the transposition of the cancelled letter when the marked connector is involved, exactly as following that strand through its first crossing prescribes.

In particular, for a factor $\gamma\eta\gamma^{-1}$ with a marked component, lift the cyclic column rotation moving γ from the front to the end by moving the protected H block aside using its exact frame. The word becomes $\eta\gamma^{-1}\gamma$, and the mark is at the interface of the inverse words. Cancel the innermost inverse pair, using the marked rule when its rank meets the mark and the unmarked rule otherwise. Reframe after each cancellation. Induction removes all of $\gamma^{-1}\gamma$ and leaves H attached to the permutation-tracked component of η. The endpoint is checked by the explicit port incidences above. No assertion that unmarked isotopy implies marked grid equivalence is used.

More explicitly, if γ has successive generator indices $i_1, \dots, i_m$, the central pairs cancel in that

order. A starting mark k ends at $s_{i_m} \cdots s_{i_1}(k) = \pi(\gamma)^{-1}(k)$, with the composition convention of the brief. Since $\pi(\gamma\eta\gamma^{-1}) = \pi(\gamma)\pi(\eta)\pi(\gamma)^{-1}$, membership in the nontrivial cycle of the conjugate implies that this ending mark belongs to the nontrivial cycle of η.

In the application the marked component belongs to the nontrivial permutation cycle of the factor. Conjugation and the successive interface cancellations preserve that property for the tracked component. Its surviving rank therefore cannot become unused. Only an unmarked rank can produce an auxiliary 2×2 block in these steps, exactly as recorded in the two cancellation cases.

All new or removed O markings in a boundary absorption lie on the marked component. Shared-endpoint exchanges involve the same component at their shared corner. Give a new O the colour of that component. Strict exchanges retain every label, and the stabilization variable identifications only identify equal colours. Therefore the entire marked construction is natural for the component-colour derivative operators by the differentiated domain identities already proved.

□

9 A signed-factorization scissor construction

Write $r = N + B$, where B is the total smooth ramification order, with each smooth local Coxeter word split into its positive-generator factors. Let a_l be the length of the nontrivial permutation cycle of the lth monodromy factor: $a_l = a_j$ at a singular point, and $a_l = 2$ at a smooth simple branch point.

Lemma 13 (Exponent count; uses the word lengths and invariant identities)**.**

Proposed name: `exteriorRamificationCount`.

Suppose the outer event word has an Artin-group factorization

$$w_{\mathrm{out}} = q_1 \cdots q_r, \qquad q_l = \gamma_l \eta_l \gamma_l^{-1},$$

where the η_l are the N local positive words and B positive generators. Then

$$B = S - \sum_{j=1}^{N} a_j + N - 1 + \mu(V).$$

Proof. The exponent-sum homomorphism sends an Artin generator to 1 and is invariant under conjugation. Hence the supplied length identities give

$$2\delta(W) + S - b = \sum_j (2\delta_j + a_j - 1) + B.$$

Substitute $\delta(W) - \sum_j \delta_j = \delta(V)$ and $2\delta(V) = \mu(V) + b - 1$. □

Lemma 14 (A common strand; new combinatorial)**.**

Proposed name: `commonStrandHurwitz`.

Let $c_1, \ldots, c_r$ be nontrivial cycles on a finite set, and suppose the group they generate is transitive. Fix a letter o. A finite algorithm of Hurwitz moves transforms the tuple into one in which every cycle moves o. The algorithm lifts to a tuple of arbitrary braid factors having these permutations, preserving its product and the individual conjugacy classes.

Proof. Call a factor good when its support contains o. At least one factor is good. If some are bad, transitivity of the support hypergraph provides a bad cycle c whose support meets that of a good cycle g. Move c next to g, keeping c itself unchanged: moving c past a factor h replaces h

by chc^{-1} or $c^{-1}hc$. Since $c(o) = o$, this does not change whether h is good. If necessary use one further Hurwitz move to put a good factor first.

For $H(g, c) = (gcg^{-1}, g)$, direct multiplication gives

$$H^{2k}(g, c) = ((gc)^k g(gc)^{-k}, (gc)^k c(gc)^{-k}).$$

The orbit of o under gc meets $\mathrm{supp}(g) \cap \mathrm{supp}(c)$. Indeed, until that orbit first enters $\mathrm{supp}(c)$, the map gc acts as g. A full g-cycle visits the nonempty intersection; its first visit to the support of c lies in the intersection. Choose an integer k such that $(gc)^{-k}(o)$ belongs to that intersection. After H^{2k} both factors are good. The number of good factors has increased, so repetition terminates. All the displayed formulas are group identities and thus apply to braid factors as well. □

Lemma 15 (Prescribed factor order; new combinatorial)**.**

Proposed name: `orderedCommonStrandHurwitz`.

Suppose every cycle in a tuple of nontrivial cycles moves a distinguished letter o. Any prescribed order of the labelled conjugacy classes can be obtained by a finite sequence of Hurwitz moves such that every cycle in the final tuple still moves o. The same algorithm applies to braid factors, preserves their product, and keeps each factor in its labelled conjugacy class.

Proof. It suffices to exchange two adjacent labelled factors g, c while retaining the distinguished letter in both final supports. Put $P = gc$. The previous Hurwitz formula gives

$$H^{2k+1}(g, c) = (P^k gcg^{-1}P^{-k},\ P^k gP^{-k}).$$

The first entry is a conjugate of c and the second a conjugate of g, so their labels are exchanged. Both entries move o precisely when

$$x_k = P^{-k}(o) \in \mathrm{supp}(g) \cap g(\mathrm{supp}(c)).$$

There is a nonnegative k with this property. Start with $x_0 = o \in \mathrm{supp}(g)$. As long as $x_k \notin g(\mathrm{supp}(c))$, the point $g^{-1}(x_k)$ is fixed by c, and hence

$$x_{k+1} = c^{-1}g^{-1}(x_k) = g^{-1}(x_k).$$

Thus, until the desired intersection is reached, the sequence follows the backward g-cycle through o. That cycle reaches $g(o)$ in at most $|\mathrm{supp}(g)| - 1$ steps, and $g(o) \in \mathrm{supp}(g) \cap g(\mathrm{supp}(c))$ because c moves o. This proves termination and gives an explicit finite search for k.

Use this adjacent exchange in an ordinary bubble sort of the factor labels. After each completed exchange all factors again move o; the number of label inversions strictly decreases. The other factors are untouched, and the displayed Hurwitz identities preserve the product at every step. All identities hold in the braid group as well, so the same signed-word operations prove the braid assertion. In the application, order the singular factors by $Q_1, \ldots, Q_N$ first, followed by the simple factors in any fixed order. □

Lemma 16 (Star scissors and the incidence forest; new combinatorial)**.**

Proposed name: `scissorIncidenceForest`.

Suppose $r \geq 1$ factors on S strands are conjugates of words with one nontrivial cycle of length $a_l \geq 2$, their permutations are transitive, and every such cycle contains strand 0. Pad each factor as specified below. The star rule gives $s = (S-1)(r-1)$ adjacent-row scissors and a final grid with one main component and $U = \sum_l (S - a_l)$ auxiliary components. Exactly U specified scissors can use the O-switch map followed by a strict row commutation; all others use the X identity. The endpoints are the same for either choice. The incidence matrix of these U O maps, with the main-component row removed, has determinant one. Moreover $s - U = \sum_l a_l - r - S + 1$.

Proof. **9.1 Signed word grids and the scissor switches**

Use the signed three-column convention of Section 2. For both signs the integer-rank incoming corner precedes the outgoing corner of that occurrence. Consequently, at any fixed integer rank, the arcs between successive occurrences have pairwise disjoint column intervals, except for the unique closing arc, whose interval contains all the others. Thus all tie orders at an integer rank are related by ordinary, four-distinct-endpoint row commutations. Half-integer arcs of distinct letters have disjoint intervals.

Pad each factor on both ends by an identity word $\rho\rho^{-1}$, where $\rho = \sigma_0 \cdots \sigma_{S-2}$. Every rank then occurs, every factor begins with $+\sigma_0$, and every factor ends with σ_0^{-1}. Padding does not alter the factor or its permutation.

Fix a rank $k \geq 1$. Let p_l be the last outgoing k-corner of factor l and q_l its first incoming k-corner. In the concatenation, let R_l be the row with O at p_l and X at q_{l+1}, indices taken modulo r. Perform the star sequence

$$(R_1, R_2),\ (R_1, R_3), \ldots, (R_1, R_r),$$

where a pair means exchanging its two X entries. After the lth stage, factors $2, \ldots, l$ close separately at this rank, while the row R_1 connects p_1 to q_{l+1}. At the last stage factor 1 closes as well. All row intervals at this rank remain pairwise disjoint or nested: the active interval spans the consecutive factors already processed, the newly closed factor intervals lie inside it, the remaining short connectors are disjoint, and the original last connector encloses the rest. Therefore each requested pair can be brought to adjacent rows by ordinary row commutations. None of the swaps creates an $O = X$ collision, since the incoming and outgoing corners of an occurrence are distinct columns.

Leave rank 0 unchanged. The total number of switches is

$$s = (S-1)(r-1).$$

An adjacent-row X switch has the same explicit chain map and filtration bound as an adjacent-column X switch: the map is the identity on the O-complex, and transposing both coordinates in the J-formula gives the same M_X difference ± 1. This proves shifts $(0, 1/2)$ and degree 0.

The O replacement at a specified scissor. Bring R_1, R_l into adjacent row positions $u, u+1$, in that order. Let their O columns be $a = p_1 < b = p_l$, and let their X columns before the switch be the current q_l, q_{l+1}. All four columns are distinct, because every factor has the stated padding. Transpose the adjacent-column O formula. It is the explicit state map

$$F_{\mathrm{row}}(x) = T^{\mathbf{1}_{\{a < x^{-1}(u+1) \leq b\}}} x$$

to the grid obtained by exchanging the O entries of these rows. It has degree -1 and shifts $(0, -1/2)$. In that grid the two row intervals are $p_l - q_l$ and $p_1 - q_{l+1}$. They are disjoint or nested, with four distinct endpoints: the first closes factor l and the second spans the active sequence of factors. Commute the two entire rows. Their O entries and labels return to their original outgoing factor columns, while their X entries have been exchanged. This is exactly the X-scissor endpoint. The row-commutation map after F_{row} gives the selected map. The detecting K of Lemma 5, followed by that same commutation, retains its canonical-class and derivative identities. Thus the incidence formula applies without any new local map assertion.

9.2 The explicit forest for O-switch selection

After common-strand normalization, every factor has just one nontrivial cycle, and that cycle contains 0. The unchanged rank-0 connections join all these components into one main component.

Every fixed rank of a factor gives one auxiliary component, reduced below to an explicit unknot block. Hence their number is

$$U = \sum_{j=1}^{N} (S - a_j) + B(S - 2).$$

At rank $k \geq 1$, the scissor edges form the star with center factor 1 and leaves $2, \ldots, r$. Its vertex at factor l is main if k belongs to that factor's nontrivial cycle, and is an auxiliary component otherwise. Transitivity ensures at least one main vertex at each rank.

If factor 1 is main, select the edges to precisely the auxiliary leaves. If it is auxiliary, let l_* be the first main leaf; select $(1, l_*)$ and the edges from 1 to every other auxiliary leaf. This is a tree rooted at the main component, with exactly one edge per auxiliary vertex. Taking all ranks gives U selected edges. The reduced incidence matrix is square with determinant 1 over $\mathbb{F}_2$: remove leaves successively, expanding along their unique incident edge. In particular this selection is suitable for an incidence-determinant formulation of the O-switch/death class calculation.

Finally,

$$s - U = (S - 1)(N + B - 1) - \sum_j (S - a_j) - B(S - 2) = \sum_j a_j + B - N - S + 1 = \mu(V).$$

□

10 Exact seam coordinates and successive factor cleanup

Lemma 17 (Exact seam coordinates and factor cleanup; new combinatorial)**.**

Proposed name: `cyclicSeamCoordinates`.

Proposed name: `seamFactorCleanup`.

Assume the scissor factors are $\gamma_l \eta_l \gamma_l^{-1}$, where η_l is supported on its a_l active ranks, contains every active generator, and has one full a_l-cycle, as in the analytic factorization. The preparation at a specified arc below gives a triply prepared grid by one X stabilization and cyclic permutations. On triply prepared factors the seam operation is strictly associative, and the displayed cyclic block rule exposes any chosen factor by explicit ordinary moves. The raw scissor endpoint is that cyclic block grid. Successive marked cancellations preserve an exact iterated-seam invariant and reduce all factors to framed core word grids with their unused-rank blocks. Applying the proved block normalizations and singleton contractions deletes exactly those auxiliary rows and columns and leaves the exact seam of the framed active core grids.

Proof. Call a grid D of size $n \geq 2$ *triply prepared* if

$$X_D(0) = 0, \qquad O_D(n-1) = 0, \qquad X_D(n-1) = n - 1. \tag{20}$$

These conditions imply all the preparation conditions required of both the left and the right factor of the supplied binary seam. In this section every iterated seam of triply prepared factors uses the identity row rotation at every stage.

10.1 Preparation at a specified arc

Let an arc of a grid G of size $g \geq 2$ run from $O(P, b)$ to $X(Z, b)$. X-stabilize at the latter marking. Insert a column W immediately before Z in the column circle, and insert a row t immediately after b in the row circle, with the three markings

$$O(W, t), \qquad X(W, b), \qquad X(Z, t), \tag{21}$$

the old $X(Z, b)$ having been removed. Cut the column circle immediately before Z, and cut the row circle immediately before t. Then Z is first, W last, t bottom, and b top. The resulting grid, denoted $D(G, b)$, is triply prepared and has size $g+1$. Formula (21) is an ordinary X stabilization; the two choices of cuts are cyclic permutations. The new O has the colour of the component containing the specified arc. No component is created or merged.

10.2 Strict associativity and a cyclic block formula

Let $D_1, \dots, D_r$ be triply prepared, with sizes n_l, and set

$$e_l = n_l - 1, \qquad c_l = \sum_{i<l} e_i, \qquad d = \sum_{l=1}^{r} e_l.$$

Their iterated seam T has size $d+1$. Substitution in the binary seam formula shows that T is itself triply prepared. It also shows that the seam operation is strictly associative on these data: both parenthesizations have the same O and X in every column, with offsets given by the corresponding sums of the e_l. In particular, no unmentioned grid move or row rotation is concealed in reassociation. Explicitly, for $0 \le j < e_l$ and $Y \in \{O, X\}$,

$$Y_T(c_l + j) = c_l + Y_{D_l}(j), \qquad O_T(d) = 0, \quad X_T(d) = d.$$

This formula follows by induction from the binary formula, because the unique zero-row O of each right factor lies in its last column and its last X is in its top row. It proves the associativity and preparation assertions directly.

Suppose $r \ge 2$. Let $M = d$ be the last column of T and $A = 0$ its first column. Across the toroidal last/first column boundary and the top/bottom row boundary there are exactly the three markings

$$X(M, d), \qquad O(M, 0), \qquad X(A, 0).$$

The fourth corner (A, d) is empty. Indeed, the X of A is in row 0, while its O is in the first factor's row block, hence in a row at most $e_1 < d$. X-destabilize this block, deleting M and row 0 and replacing its X markings by $X(A, d)$ before renumbering. A cyclic cut makes this an ordinary adjacent-row, adjacent-column destabilization if the convention requires a planar 2×2 block.

Denote the resulting size-d grid by $C(D_1, \dots, D_r)$. Its columns and rows are divided into blocks of sizes e_l. For $0 \le j < e_l$ its complete marking formula is

$$\begin{aligned} O_C(c_l + j) &= c_l + O_{D_l}(j) - 1, \\ X_C(c_l + j) &= \begin{cases} c_l - 1 \pmod d, & j = 0, \\ c_l + X_{D_l}(j) - 1, & j > 0. \end{cases} \end{aligned} \tag{22}$$

All values on the first line lie in the lth row block, since the only O in row 0 of D_l is in its omitted last column. The same is true of the second line for $j > 0$. The first X of a block lies in the top row of the preceding block, with the blocks read cyclically. This is the entire effect of the destabilization.

The formula is literally cyclically symmetric. To put factor D_l first, cyclically rotate the column blocks by c_l and the row blocks by the same amount. These are permitted cyclic column and row permutations. Apply the inverse X stabilization at the first-column X in the top row: add a last column M and a bottom row 0, place $O(M, 0), X(M, \text{top}), X(0, 0)$, and remove the old first-column/top-row X. One obtains exactly

$$\Sigma(D_l, D_{l+1}, \dots, D_r, D_1, \dots, D_{l-1}),$$

where the displayed expression is strictly associative. Thus any factor can be exposed as the left factor of a binary seam whose right factor is the contiguous seam of all the others. This conclusion is proved by marking coordinates, without a commutativity assertion for connected sums.

The formula includes factors of size 2: then $e_l = 1$ and the $j > 0$ case for that block is simply absent. Since $r \geq 2$, $d \geq 2$, so the temporary destabilized grid never has size 1. When $r = 1$ there is no cyclic factor permutation to perform, and the operation is omitted; its fourth corner need not be empty in that case.

10.3 The raw scissor endpoint is already this cyclic grid

Let $q_1, \ldots, q_r$ be the padded signed factors in the scissor construction, and let G_l be their standalone word grids. At each factor the first letter is σ_0 and the last letter is σ_0^{-1}. Thus its first column A_l is the first incoming rank-0 corner, its last column B_l is the last outgoing rank-0 corner, and its closing rank-0 row b_l runs $O(B_l, b_l) \to X(A_l, b_l)$.

Apply the preparation formula (21) to the standalone grid G_l at this arc and call the result D_l. This defines D_l explicitly; it is not an assertion that unrelated factor moves lift through a seam. Its nonlast columns are exactly the columns of G_l, in their original cyclic order, and its nonbottom rows are exactly the rows of G_l, cut just after b_l, so that b_l is last. Its omitted last column and bottom row are precisely the added frame.

After the scissor switches at every rank other than 0, the raw endpoint has these same column blocks. Every row is internal to one block except the rank-0 rows

$$R_l : O(B_l) \longrightarrow X(A_{l+1}), \qquad l \pmod r.$$

Assign R_l to row block l. Within each block put the rows in the cyclic order of G_l starting immediately after its closing rank-0 row; put R_l last. These row permutations are realized as follows. Rows internal to different column blocks have disjoint intervals. For $l < r$, R_l has consecutive column endpoints B_l, A_{l+1}; R_r is the closing interval containing every other column. Hence every pair of rows with different block labels has disjoint or nested intervals, or shares an endpoint. Every needed adjacent exchange is therefore either a strict noninterleaving commutation or the explicit shared-endpoint replacement already proved.

To implement independent cyclic cuts, gather one row colour into a consecutive block by these exchanges. Move its first row globally around the row circle, then exchange it back past all other colours until it is last in its own block. Repeating realizes any cyclic rotation of that colour while preserving all other cyclic orders. After choosing the required cyclic order for every colour, use a stable sort of the colour labels, which exchanges only different colours. This is a finite algorithm producing the stated row order.

The raw scissor endpoint in this order is exactly $C(D_1, \ldots, D_r)$ by (22): its only changed X is the first one in each block, now on the preceding block's last row. Consequently a single global X stabilization, the inverse of the destabilization above, gives the exact iterated seam of the triply prepared D_l. This preparation changes neither the saddle count nor any component colour. The added O is on the main component. For $r = 1$ the raw endpoint is simply G_1, and this same stated stabilization is exactly $G_1 \to D(G_1, b_1)$. It does not rely on a preceding cyclic-seam destabilization.

10.4 The invariant under marked inverse-pair cancellation

Maintain the following exact invariant: the current grid is the strict iterated seam of triply prepared factor grids, one for each monodromy factor; a factor consists of its current marked word part and the explicitly retained 2×2 blocks of its unused ranks, each with its two rows and its two columns adjacent. All such auxiliary components keep their prescribed colours. The cyclic block rule exposes any desired factor D first. By strict associativity the resulting grid is exactly

$$\Sigma(D, H),$$

where H is the contiguous seam of the other factors. For $r \geq 2$, H has size at least 2, is triply prepared, and its complete marking data will be protected during the following operation.

Here $D = D(G, b)$ is the one-stabilization preparation of its underlying marked word part, including any retained auxiliary blocks. The binary seam formula already gives its exact framed insertion form: the word column Z is retained with X in the bottom row t, the first H column has X in the top base row b, and the last H column has O in t. Its internal H rows lie above b. No additional stabilization to restore the virtual frame column is performed. Cancel the padding and the conjugator of the exposed factor by the marked inverse-pair theorem, using unmarked inverse-pair cancellation whenever the current mark does not meet the two affected ranks. At every marked cancellation the protected H block retains its columns, internal rows, their orders, and every internal marking. Only its incoming and outgoing port rows change. The local theorem restores the exact frame after each cancellation. For a factor $\gamma\eta\gamma^{-1}$, the successive transpositions track the mark into the nontrivial permutation cycle of η. Therefore the marked rank never becomes unused. The unused ranks are exactly the $S - a_l$ fixed strands of that factor, and each is retained as one explicit isolated 2×2 grid.

It remains to check that the output frame recovers the same global invariant, rather than merely representing an attached summand. Let the resulting base word part have the arc $O(P, b) \to X(Z, b)$. Its framed insertion has H input $X(H_{\text{first}}, b)$ and H output $O(H_{\text{last}}, t)$, while the retained base column Z has X in row t. Introduce the *virtual* last column W with $O(W, t), X(W, b)$. First use the frame's proved column-exchange rule to move the entire H block immediately before the retained base column Z, preserving all internal and exterior column orders. Then cut immediately before Z, so the base columns come first and the H block last. With row cut t first and b last among the base rows, the virtual base grid is exactly $D(G, b)$ from (21). The virtual column is precisely the column deleted by the seam formula. Substitution in that formula recovers all actual markings of the frame, including $X(H_{\text{first}}, b)$ and $O(H_{\text{last}}, t)$. Thus the actual output grid is literally

$$\Sigma(D(G, b), H).$$

The exposed factor has been replaced by another triply prepared one, and every other factor is unchanged. This proves the induction invariant and permits the next factor to be exposed by the cyclic block rule. No general statement that an unmarked equivalence lifts through a seam is needed.

Postpone all auxiliary contractions until every factor has been cleaned. An auxiliary 2×2 block can then be placed at the prescribed location by ordinary moves. First move its two columns together to the desired column position: their row endpoints are the same two adjacent rows, and no exterior column uses either row, so all these exchanges are strictly noninterleaving. With its columns adjacent, move its two rows to the desired row position; the same argument applies with coordinates transposed. The block has no shared marking with the main component. If necessary exchange its two rows by the proved shared-endpoint rule to give the four-mark pattern of Proposition 9.

For each block first apply that proposition's explicit degree-zero, shift-zero collapse to a doubled singleton. These maps preserve the canonical class and intertwine its component-colour derivative. Only after these collapses apply the degree-$+1$, shift-$(0, 1/2)$ singleton contractions of Lemma 8, using the persistent main component as root. These are the U contractions in the incidence-determinant calculation. A four-mark block is never treated as an ordinary three-corner destabilization. Intermediate doubled singletons are auxiliary marking pairs with the explicitly defined complexes and maps of those two results.

Their exact target formulas delete only the selected auxiliary rows and columns, retaining all others in their original relative order. In particular the first-column X, bottom O, and last-column top X of every main frame remain. At the end the actual grid is exactly the seam of the same factor frames with their auxiliary blocks removed. The persistent main component has a row with

distinct O and X throughout, as required for the contractions. This proves the target claim at the level of marking permutations as well as the required colour naturality.

After all factors have been processed and the prescribed U blocks have died, each remaining factor is the frame preparation of wordGrid(η_l) at a rank in its unique active cycle. Its size is at least 3, since the unframed word grid has size at least 2. Reversing its one frame stabilization, together with its explicit cyclic cuts, gives an ordinary-move equivalence with wordGrid(η_l). Thus the preparation condition and the required `MovesTo` condition are both verified without any further rank normalization.

□

11 Removing the simple factors and fixing the target

Lemma 18 (Simple-factor projection and exact target; new combinatorial, uses the binary seam comparison)**.**

Proposed name: `simpleFactorProjection`.

Suppose the cleaned factors have been ordered with the singular factors first in their prescribed order and the simple factors last. The functional (23) *on* $C(\mathrm{wordGrid}(\sigma_0))$ *is a* Λ*-linear degree-zero chain map to* Λ *with shifts* $(0,0)$ *and sends the canonical class to* 1*. Consequently the seam of the cleaned singular and simple factors admits a degree-zero, shift-zero, canonical-class-preserving map to the prescribed iterated seam of the singular factors alone.*

Proof. All ordinary moves and all prescribed auxiliary deaths in the preceding section occur before using a seam–tensor comparison. Thus the component-colour derivative proof of the O-switch/death canonical-class identity applies to the explicit marking pairs and their specified maps throughout. At that stage the canonical-class calculation is the one completed in Section 12. The following operations add no further degree or filtration shift.

For a smooth-ramification factor, its active word is a positive generator on two strands. Its three-by-three word grid has

$$O = (1,0,2), \qquad X = (0,2,1).$$

Here a state such as 021 denotes the values of its row permutation in increasing column order. Direct enumeration of the two possible empty rectangles for each transposition gives

x	∂x
012	$021 + 102 + T\,210$
120	$021 + 102 + T\,210$
201	$021 + 102 + T\,210$
021	0
102	0
210	0.

The Maslov and Alexander values of the states used below are

$$(M_O, A)(021) = (-2,-1), \qquad (M_O, A)(102) = (-2,-2), \qquad (M_O, A)(210) = (0,0).$$

With Λ having zero differential and its usual monomial degree and bifiltration, define the Λ-linear map

$$\epsilon(210) = 1, \qquad \epsilon(021) = T, \qquad \epsilon(012) = \epsilon(102) = \epsilon(120) = \epsilon(201) = 0. \tag{23}$$

The displayed differential table gives $\epsilon\partial = T + T = 0$. The two nonzero images have exactly the degrees of their inputs. Their filtration coordinates satisfy

$$(0,0) = (0, A(210)), \qquad (-1,-1) \le (0, A(021)) = (0,-1).$$

Multiplication by T^k translates both input and output coordinates by $(-k,-k)$, proving shifts $(0,0)$ for every monomial. The degree-zero cycle 210 lies in the polynomial subcomplex, and (23) proves its class nonzero. Hence it is the supplied canonical class and is sent to 1.

Before the scissor construction, use Lemma 15 to order the labelled singular factors as $Q_1,\dots,Q_N$, followed by the B simple factors, while keeping strand 0 in every nontrivial factor cycle. The subsequent factor exposures change this order only by cyclic rotation: each uses the cyclic block rule, and marked cancellation changes only the exposed factor. After cleanup and the auxiliary contractions, apply that same cyclic block rule to put the factor labelled Q_1 first. This is possible because $N \geq 1$. The actual grid is now the exact seam of the cleaned singular factors in their prescribed order, followed by the cleaned simple factors. All these ordering operations are ordinary moves of degree and shift zero.

For a cleaned simple factor H, define

$$\epsilon_H = \epsilon \circ M_H : C(H) \longrightarrow \Lambda,$$

where M_H reverses its one frame stabilization and cyclic cuts to the three-by-three word grid. This map is Λ-linear. Indeed, the ordinary maps chosen here are the explicitly proved polynomial domain maps: a commutation is its pentagon count, the primitive stabilization directions are the incoming and outgoing domain counts, and the other corners use their finite corner reductions. Each is a sum of state terms with powers of T, extended Λ-linearly. Cyclic row and column permutations use their literal state bijections and preserve rectangle weights; these too are Λ-linear. The earlier proofs give their degree, shift, and class properties. Consequently ϵ_H has degree zero, shifts $(0,0)$, and sends $\mathrm{can}(H)$ to 1.

Remove the last simple factor one at a time. By the strict seam formula the current grid is $\Sigma(G,H)$, where H is that last simple factor and G is the exact seam of the remaining prefix. The prefix always includes the $N \geq 1$ singular factors, so it is a valid grid of size at least two. Apply the supplied *binary* comparison, followed by

$$C(\Sigma(G,H)) \xrightarrow{\Theta_{G,H}} C(G) \otimes_\Lambda C(H) \xrightarrow{\mathrm{id}\otimes\epsilon_H} C(G) \otimes_\Lambda \Lambda \cong C(G). \tag{24}$$

The second arrow is well defined because ϵ_H is Λ-linear. It is a chain map of degree zero and shifts $(0,0)$: if a monomial term of $\epsilon_H(y)$ is T^m, its shift bounds say $m \geq 0$ and $-m \leq A_H(y)$, so the image of $T^k(x \otimes y)$ has coordinates

$$(-k-m, A_G(x)-k-m) \leq (-k, A_G(x)+A_H(y)-k).$$

The canonical-class identity of the binary comparison, followed by $\epsilon_H\,\mathrm{can}(H) = 1$, proves the identity for (24).

Repeat this operation exactly B times. Its exact target after each step is the prefix grid G, so no comparison is tensored with an identity map and no inverse or iterated tensor comparison is used. In particular no Λ-linearity of the supplied $\Theta_{G,H}$ is assumed. The final target is literally the seam of the retained singular factors $D_1,\dots,D_N$ in their required order, with identity row rotations. Every step has degree zero, shifts $(0,0)$, and preserves the canonical class. If $B = 0$ this entire removal operation is the identity.

The original scissor uses s saddles, of which U are O switches; the other $s-U$ are X switches. The O switches and their U deaths cancel in degree and filtration shift. Since the exact count proved earlier gives $s - U = \mu(V)$, the resulting map has degree zero and shifts $(0,\mu(V)/2)$. All size bounds required in Statement E hold, and every local factor is related by ordinary moves to its supplied positive word grid.

If $r = 1$, transitivity forces the sole nontrivial cycle to contain all S ranks. Since there is at least one singularity, this sole factor is a singularity factor. Thus $U = s = 0$ and the exact count gives $\mu(V) = 0$. There is no scissor and no factor permutation. Apply standalone cyclic word

rotation and the unmarked free-cancellation rule to reach its local positive word grid, then the one stabilization (21) gives its prepared target. This separately covers the only case in which the cyclic seam destabilization was not used.

□

12 Construction of the complete map and proof of Statement E

Proof of Theorem 1. Choose the assembly supplied by the brief. Apply the finite-projection and event-factorization lemma to choose a sufficiently small allowed level and a projection transverse at every central and local branch. It gives the central positive word $\alpha = \mathbf{d}_S\mathbf{w}$, the local positive words β_j, and a finite derivation, using signed Artin relations and conjugacy, of

$$\alpha \ \sim \ q_1 \cdots q_r, \qquad q_l = \gamma_l \eta_l \gamma_l^{-1}, \qquad r = N + B.$$

If the initial comparison is by whole-word conjugacy, absorb its conjugator c into every γ_l. Indeed $c(q_1 \cdots q_r)c^{-1} = \prod_l (cq_lc^{-1})$, with a finite free derivation obtained by inserting $c^{-1}c$ between consecutive factors. The guarded word normalization can therefore use only local Artin and free relations in its suffix. Each η_l is a local β_j, supported on its first a_j strands, or σ_0 on its first two strands. A cluster originally in another consecutive interval is put in this form by the explicit identity $\mathrm{sh}_k(\eta) = \rho_S^k \eta \rho_S^{-k}$. The factor permutations generate a transitive group on S points. In particular every local degree is at most S, since it is a cluster in an S-point regular fibre.

Apply the finite Hurwitz algorithm with distinguished strand 0. It preserves the product and the individual conjugacy classes and puts strand 0 in every nontrivial factor cycle. Remember the original singularity index attached to each local factor. The odd-Hurwitz interchange rule then orders the factors as the singular factors $1, \ldots, N$, followed by all the simple factors, with strand 0 still in every nontrivial cycle after each completed interchange. Pad each factor on both ends by $p = \rho_S \rho_S^{-1}$. Since $p^{-1} = p$ as a literal signed word,

$$p\gamma_l \eta_l \gamma_l^{-1} p = (p\gamma_l)\eta_l(p\gamma_l)^{-1}.$$

The signed-word rules, with the external guard described there, give an explicit ordinary-move map from the specified native source grid to the grid of this concatenation. It has degree and shift zero and sends the canonical class to the canonical class.

Perform the star scissors at ranks $1, \ldots, S-1$. There are $s = (S-1)(r-1)$ operations. Select the U edges of the explicitly specified rooted forest, where $U = \sum_j (S - a_j) + B(S-2)$. At every unselected edge use the adjacent-row X identity. At every selected edge use the transposed diagonal O-switch map followed by the specified noninterleaving whole-row commutation. Thus both choices have exactly the same endpoint markings. The selected choice has degree -1 and shifts $(0, -1/2)$, while the unselected choice has degree zero and shifts $(0, 1/2)$.

Colour the O markings by their components at the completed scissor endpoint: colour 0 is the single main component, and the other U colours are its auxiliary components. Transport these labels back through the scissors. This is legitimate without any monochromatic assumption on an earlier component. Every intervening row exchange during the scissors has four distinct endpoint columns, so it uses a labelled commutation and identifies no O variables. An X switch leaves the O differential and its labels unchanged. For a selected O switch the two final endpoint colours are exactly the two vertices of its forest edge. The final incidence matrix, with its root row removed, therefore has determinant one.

Use the explicit cyclic-grid stabilization to obtain the exact seam of triply prepared factors. Expose one factor at a time by the cyclic seam formula. Its padding and conjugator are removed by the marked inverse-pair rules while the complementary factor grid is retained as the protected block. The exact frame is restored after every cancellation and the resulting factor is again triply

prepared. Thus this is an actual finite sequence on the whole grid. The tracked rank is in the core's nontrivial cycle. Precisely its $S - a_l$ fixed ranks become four-mark 2×2 components; the marked rank never disappears. Complete this procedure for every factor.

For a retained four-mark block with the opposite O diagonal, exchange its two adjacent rows by the explicit shared-endpoint rule, so that it has the displayed collapse pattern. Normalize each of the U four-mark blocks to a doubled singleton by the explicit degree-zero, shift-zero, colour-natural map of Proposition 9. These are algebraic normalizations, with their full formulas proved above. They cannot be replaced by three-corner destabilizations. All ordinary moves and these normalizations preserve the canonical class and intertwine every component-colour derivative on homology. Denote their composite, following the scissors, by P.

Finally remove the U doubled singletons by the explicit contraction maps, always taking a surviving O of the main component as root. For the singleton of colour c its contraction is, on homology, $\rho_c \Psi_c$, with ρ_c of degree zero and second shift $1/2$, preserving the canonical class. A derivative of another dying colour commutes past ρ_c; the possible root-substitution correction contains Ψ_c^2 and is zero. Hence the induced map of all contractions is the ordered product of the ρ_c's following the product of all U dying-colour derivatives. Pass those derivatives back through P and use the determinant identity (13). Since the reduced forest incidence matrix has determinant one, the result before the ρ_c's is exactly the canonical class. Each ρ_c preserves that class. This proves nonvanishing and, by Lemma 2, the required canonical-class identity for the entire map so far.

The target markings at this stage are literally the seam of the cleaned triply prepared factor grids. Deleting a singleton deletes only its row and column and preserves the order of all retained rows and columns. The auxiliary blocks occupy no main frame port, so these exact deletions commute with the displayed seam marking formula. Every cleaned factor is the one-frame preparation of the word grid of its η_l, at its tracked active arc.

The cyclic exposures preserve the cyclic order of all factors. Use the explicit cyclic seam rule to restore the chosen first factor, so that the singular factors are again $1, \dots, N$ and the simple factors are last. If a simple factor remains, write the actual grid as $\Sigma(G, H)$, where H is its last prepared simple factor. Apply the supplied *binary* comparison

$$C(\Sigma(G, H)) \longrightarrow C(G) \otimes_\Lambda C(H).$$

Reverse the frame preparation of H and apply (23), obtaining a Λ-linear map $\epsilon_H : C(H) \to \Lambda$ of degree and shift zero. Consequently $\mathrm{id}_{C(G)} \otimes_\Lambda \epsilon_H$ is well defined and gives a map to $C(G)$. This composition uses no Λ-linearity of the supplied binary comparison. It preserves the canonical class. Repeat for the B final simple factors. No iterated tensoring of that comparison, or derivative naturality for it, is used.

All these maps have degree zero and shifts $(0, 0)$. Put D_j equal to the retained prepared grid of β_j. It is related to wordGrid(β_j) by the explicitly given moves. The exact final grid is iteratedSeamGrid(D), with every row rotation equal to the identity.

The degree of the composite is $-U + U = 0$. Its first shift is zero. Its second shift is

$$\frac{s - U}{2} - \frac{U}{2} + \frac{U}{2} = \frac{s - U}{2} = \frac{\mu(V)}{2}.$$

Every equality uses the explicitly verified bounds for its maps; in particular the four-mark normalizations contribute zero. The available local words give $a'_j = a_j \in [2, t_J]$ and all their required cycle, generator, and length properties. The central word has the required properties by (I1)–(I2).

If $r = 1$, the separate construction in the endpoint lemma uses standalone conjugator cancellation and one frame stabilization. Transitivity gives $a_1 = S$, and $N \geq 1$ gives $N = 1, B = 0$. Thus $U = s = \mu(V) = 0$, and the same conclusion follows with no scissors, auxiliary contractions, or tensor-factor deletions. This proves every assertion of Statement E. □

12.1 Complete operation count

The finite generating rules above specify every marking permutation, every adjacent pair, and every three-corner destabilization. Since this is a direct algebraic construction, its count must distinguish the maps of nonzero degree from ordinary X switches:

operation	number	degree	second shift
scissor X identity	$s - U = \mu(V)$	0	1/2
selected scissor O map	U	-1	$-1/2$
four-mark normalization	U	0	0
final singleton contraction	U	$+1$	1/2
ordinary and endpoint maps	finite	0	0

There are no births. If every four-mark normalization is expanded into its defining O switch and singleton contraction, the complete formula uses $2U$ O switches and $2U$ singleton contractions, besides the $\mu(V)$ X identities. The added pairs still have total degree and shift zero. Thus the effective X count minus birth count is exactly $\mu(V)$, and the actual shift computation is the displayed sum, not an application of the restricted M/X/B movie assertion.

All counts are explicit functions of the brief's invariants:

$$\begin{aligned} B &= S - \sum_j a_j + N - 1 + 2\delta(V) - b + 1, \\ r &= N + B, \qquad s = (S-1)(N+B-1), \\ U &= NS - \sum_j a_j + B(S-2). \end{aligned}$$

The number of zero-shift moves is finite by the stated algorithms: the finite event derivation, finite Hurwitz iteration, finite bubble sorts, the 29-step Artin rule, and finitely many inverse cancellations. No bound on that harmless count is needed.

13 A fully indexed example with $S = 4$ and $\mu = 1$

This is a numerical and combinatorial instance of the construction; no assertion is made that these data are realized by a holomorphic counterexample. Write 012 for $\sigma_0\sigma_1\sigma_2$, and take

$$\alpha = (012)^5 0 = \mathbf{d}_4\,(0120), \qquad \beta_1 = (012)^5, \qquad \eta_2 = \sigma_0.$$

The numerical data are

$$S = 4, \quad M = 2, \quad b = 2, \quad N = 1, \quad a_1 = t_J = 4, \quad \delta(V) = 1, \quad \delta(W) = 7, \quad \delta_1 = 6, \quad \mu(V) = 1.$$

They satisfy $2\delta(V) + M = S$, $\delta(W) = \delta(V) + \delta_1$, $\mu(V) = 2\delta(V) - b + 1$, and $\delta(W) \geq S(S-1)/2$. The lengths are $\ell(\alpha) = 16 = 2\delta(W) + S - b$ and $\ell(\beta_1) = 15 = 2\delta_1 + a_1 - 1$. With the stated permutation convention,

$$\pi(\beta_1) = (0\ 1\ 2\ 3), \quad \pi(\eta_2) = (0\ 1), \quad \pi(\alpha) = (0\ 2\ 3)(1).$$

Thus $B = 1$, both factor cycles contain 0, and the common-strand algorithm needs no Hurwitz move in this example.

For the source and the unprepared local grid, order equal-height arcs by their outgoing column. The columns in the following tables are zero-indexed; each pair of rows $O(c), X(c)$ specifies the complete native tuple (O, X).

Source grid: size 48**.**

c	0	1	2	3	4	5	6	7	8	9	10	11
$O(c)$	6	0	12	23	13	28	38	29	43	7	1	14
$X(c)$	5	22	6	12	37	23	28	47	38	0	13	7
c	12	13	14	15	16	17	18	19	20	21	22	23
$O(c)$	24	15	30	39	31	44	8	2	16	25	17	32
$X(c)$	14	29	24	30	43	39	1	15	8	16	31	25
c	24	25	26	27	28	29	30	31	32	33	34	35
$O(c)$	40	33	45	9	3	18	26	19	34	41	35	46
$X(c)$	32	44	40	2	17	9	18	33	26	34	45	41
c	36	37	38	39	40	41	42	43	44	45	46	47
$O(c)$	10	4	20	27	21	36	42	37	47	11	5	22
$X(c)$	3	19	10	20	35	27	36	46	42	4	21	11

Put

$$p = 012\,\overline{2}\,\overline{1}\,\overline{0}, \qquad q_1 = p\beta_1 p, \qquad q_2 = p0p, \qquad Q = q_1 q_2,$$

where a bar denotes a negative generator. Their lengths are $27, 13, 40$, respectively. Inserting the four displayed identity words gives an explicit free-insertion derivation from α to Q; the signed-word rules realize it by ordinary grid moves. The first factor uses columns $0, \dots, 80$, and the second uses columns $81, \dots, 119$.

At each integer rank, put all internal arcs first in outgoing-column order, then put the two connector rows R_1, R_2 last, in that order. At each half-integer height use outgoing-column order. These tie changes are the already proved ordinary row moves. The resulting complete padded grid is as follows.

Padded grid G^0: size 120.

c	0	1	2	3	4	5	6	7	8	9	10	11
$O(c)$	14	0	28	55	29	68	94	69	107	95	108	70
$X(c)$	13	54	14	28	93	55	68	119	94	107	69	95
c	12	13	14	15	16	17	18	19	20	21	22	23
$O(c)$	56	71	30	15	31	1	16	2	32	57	33	72
$X(c)$	70	29	56	30	0	15	1	31	16	32	71	57
c	24	25	26	27	28	29	30	31	32	33	34	35
$O(c)$	96	73	109	17	3	34	58	35	74	97	75	110
$X(c)$	72	108	96	2	33	17	34	73	58	74	109	97
c	36	37	38	39	40	41	42	43	44	45	46	47
$O(c)$	18	4	36	59	37	76	98	77	111	19	5	38
$X(c)$	3	35	18	36	75	59	76	110	98	4	37	19
c	48	49	50	51	52	53	54	55	56	57	58	59
$O(c)$	60	39	78	99	79	112	20	6	40	61	41	80
$X(c)$	38	77	60	78	111	99	5	39	20	40	79	61
c	60	61	62	63	64	65	66	67	68	69	70	71
$O(c)$	100	81	113	21	7	42	62	43	82	101	83	114
$X(c)$	80	112	100	6	41	21	42	81	62	82	113	101
c	72	73	74	75	76	77	78	79	80	81	82	83
$O(c)$	102	118	84	63	92	44	22	53	12	23	8	45
$X(c)$	114	83	102	84	43	63	44	7	22	12	53	23
c	84	85	86	87	88	89	90	91	92	93	94	95
$O(c)$	64	46	85	103	86	115	104	116	87	65	88	47
$X(c)$	45	92	64	85	118	103	115	86	104	87	46	65
c	96	97	98	99	100	101	102	103	104	105	106	107
$O(c)$	24	48	9	25	10	49	26	11	50	66	51	89
$X(c)$	47	8	24	9	48	25	10	49	26	50	88	66
c	108	109	110	111	112	113	114	115	116	117	118	119
$O(c)$	105	90	117	106	119	91	67	93	52	27	54	13
$X(c)$	89	116	105	117	90	106	91	51	67	52	11	27

The three scissor operations are now adjacent-row operations without any preliminary row exchanges. Both their row positions and the columns containing the exchanged entries are explicit:

rank	(R_1, R_2)	O columns	X columns	operation
1	$(53, 54)$	$(79, 118)$	$(1, 82)$	X
2	$(92, 93)$	$(76, 115)$	$(4, 85)$	O, then whole-row commutation
3	$(118, 119)$	$(73, 112)$	$(7, 88)$	O, then whole-row commutation

Here the X operation is the explicitly proved adjacent-row version: its chain map is the identity on the O complex. The listed X columns identify the two tuple entries; they are not asserted to be adjacent columns. Transposing the adjacent-column grading calculation proves its second shift $1/2$.

For exact intermediate tuples, let $\tau_{a,b}Y$ mean exchange entries a, b of a permutation tuple Y, and let $s_{u,v}Y$ mean exchange the row values u, v wherever they occur. If the displayed padded tuple is (O, X), put

$$X_1 = \tau_{1,82}X, \qquad X_2 = \tau_{4,85}X_1, \qquad X_3 = \tau_{7,88}X_2.$$

The full successive grid list, including the O-switch targets, is

$$\begin{aligned}
G^0 &= (O, X),\\
G^1 &= (O, X_1),\\
H^2 &= (\tau_{76,115}O, X_1),\\
G^2 &= (s_{92,93}\tau_{76,115}O, s_{92,93}X_1) = (O, X_2),\\
H^3 &= (\tau_{73,112}O, X_2),\\
G^3 &= (s_{118,119}\tau_{73,112}O, s_{118,119}X_2) = (O, X_3).
\end{aligned}$$

Every tuple is valid. The two whole-row commutations are strict: after the first O exchange their marking intervals are $[4, 76]$ and $[85, 115]$; after the second they are $[7, 73]$ and $[88, 112]$. Thus no shared-endpoint convention is needed here. On a state x the O-switch maps in these coordinates are explicitly

$$F_2(x) = T^{\mathbf{1}_{\{76<x^{-1}(93)\le 115\}}}x, \qquad F_3(x) = T^{\mathbf{1}_{\{73<x^{-1}(119)\le 112\}}}x.$$

These are the transposed O-switch formulas already proved, so each has degree -1 and shifts $(0, -1/2)$.

The component counts of G^0, G^1, G^2, G^3 are respectively $2, 1, 2, 3$. At G^3 the rank-0 connections form the main component; the fixed ranks $2, 3$ of q_2 give the other two components. The selected O-switch graph has one edge from the main component to each of them, and its reduced incidence matrix is I_2. The determinant pairing therefore proves nonvanishing for the two prescribed contractions. The marked cancellation algorithm removes both copies of p from each factor. It leaves the active words β_1 and σ_0, together with exactly the two retained ordinary 2×2 auxiliary blocks. Applying the proved four-mark collapse and singleton contraction to each block, and then the simple-factor functional after the seam–tensor comparison, gives the stated local target. This uses the map construction of Statement E, with its explicit degree -1 and $+1$ intermediate maps.

The count is

$$r = 2, \qquad s = (S-1)(r-1) = 3, \qquad U = (S-a_1) + B(S-2) = 2, \qquad s - U = 1 = \mu(V).$$

There is one X switch of degree 0 and second shift $1/2$, two O switches of degree -1 and second shift $-1/2$, and two prescribed auxiliary contractions of degree $+1$ and second shift $1/2$. The four-mark collapses each have degree and shift zero. Expanding their formulas adds two further O switches and two further singleton contractions; thus the completely expanded map uses four of each, still with one X switch. These added pairs cancel in both degree and shift. Consequently the complete map has degree 0, shifts $(0, 1/2)$, and sends the canonical class to the canonical class.

Finally choose the local prepared grid D_1 explicitly. In the outgoing-column tie order of wordGrid(β_1), its closing rank-0 arc is in row $b = 4$, ending in column 0. X-stabilize there by inserting a column immediately before column 0 and a row immediately after row 4, with the pattern $O(W, t), X(W, 4), X(0, t)$. Cut at old column 0 and the new row t. This produces the following size-46 tuple. Its first X and last O are in row 0, its last X is in row 45, and reversing that single stabilization proves $D_1 \sim$ wordGrid(β_1). Since $N = 1$, the target iterated seam is exactly this grid.

Target prepared grid D_1: size 46.

c	0	1	2	3	4	5	6	7	8	9	10	11
$O(c)$	1	41	6	16	7	21	31	22	36	2	42	8
$X(c)$	0	15	1	6	30	16	21	40	31	41	7	2
c	12	13	14	15	16	17	18	19	20	21	22	23
$O(c)$	17	9	23	32	24	37	3	43	10	18	11	25
$X(c)$	8	22	17	23	36	32	42	9	3	10	24	18
c	24	25	26	27	28	29	30	31	32	33	34	35
$O(c)$	33	26	38	4	44	12	19	13	27	34	28	39
$X(c)$	25	37	33	43	11	4	12	26	19	27	38	34
c	36	37	38	39	40	41	42	43	44	45		
$O(c)$	5	45	14	20	15	29	35	30	40	0		
$X(c)$	44	13	5	14	28	20	29	39	35	45		

All tables and intermediate permutations above are generated by the accompanying elementary script. It checks every permutation and O/X inequality, computes the component cycles as $X^{-1} \circ O$, verifies the displayed row operations and the strict noninterleaving inequalities, and checks the target frame coordinates. The chain-map, degree, filtration, and class claims follow from the proved general formulas, rather than from a numerical test of the 120!-state complex.

14 Exceptional cases and exact scope

Case from the brief	Treatment in the construction
One central branch, $b = 1$	Nothing in the scissor or class argument requires multiple boundary components. The formula gives $\mu(V) = 2\delta(V)$, so the X count is even as required.
One singular point, $N = 1$	There can still be $B > 0$ simple factors. The construction first handles all $r = 1 + B$ factors and then removes the simple tensor factors. The final iterated seam is just D_1. The distinct case $r = 1$ is treated explicitly in the proof.
Several tangent directions	Choose a central transverse direction first, construct the finite representative and the global section, then choose a sufficiently small allowed level. A small projection shear avoids the finite set of forbidden central and local tangent directions without changing that level. Higher ordinary ramification and repeated critical values are handled explicitly; simplicity is not assumed.
Empty **w**, ordinary S-fold central point	The word $\mathbf{d}_S$ still contains every generator, so every combinatorial rule applies. Under the full assembly identities the further condition $b = S$ gives $\mu(V) = 2\delta(V) - S + 1 = 1 - M < 0$, since $M \geq 2$; hence that entire ordinary-point specialization cannot occur for such an assembly. This elementary numerical observation is not an invocation of the conjecture.
Smallest $S = 4$	No estimate is weakened at this value. In the $\mu(V) = 1$ case there is exactly one ordinary X identity, with the O/contraction pairs having total shift zero. Section 13 gives all relevant indexed tuples for a finite example.

Case from the brief	Treatment in the construction
Missing generators	The original source contains every generator because of its full twist. The external guard retains every generator throughout initial signed-word normalization. Later unused ranks are kept as explicit two-row, two-column components of the defined signed extension and are removed only by the proved auxiliary maps.
Grid sizes and successor sizes	The positive word grids have size $3\ell \geq 3$. Every prepared factor has one additional row and column. A cyclic seam destabilization is used only with at least two factors, so its temporary size is at least two. Auxiliary contractions retain the nontrivial main factor. No size-zero or size-one complex occurs. Every final factor size is a successor as required.
$U = 0$, $B = 0$, or $s = 0$	Products over empty sets are identities and the empty incidence determinant is one. No auxiliary map is used when $U = 0$, and no functional is used when $B = 0$. The case $s = 0$ is exactly the separately handled $r = 1$ case.
Large $\sum_j a_j - (N - 1)$	The construction never assumes this quantity is at most S. Each individual $a_j \leq S$ follows from the finite projection. Any excess in the sum is absorbed by the actual count $\sum_j a_j + B - N - S + 1 = \mu(V)$.

15 Dependency ledger and references

All entries labelled new below are proved in this document. The names are proposed short names for later formalization; no claim of a completed Lean formalization is made.

Result	Proposed short name and type	Status
Nonvanishing implies the class identity	`polynomialImageNonzero`; uses canonical image	Proved
Composition along ordinary moves	`movesToNativeMap`; uses primitive moves and corner reduction	Proved
Finite projection and event factorization	`finiteProjectionEventFactorization`; new analytic	Proved
O derivatives, saddle pairing, positive-degree vanishing, incidence formula	`nativeOSaddleIncidence`; new combinatorial	Proved
Explicit incoming and outgoing polynomial projection formulas	`nativeTopDomainMaps`; new combinatorial	Proved
Colour naturality of native moves	`nativeMoveColourNaturality`; new combinatorial	Proved
Singleton contraction, transport, and substitution	`nativeAuxiliaryContraction`; new combinatorial	Proved
Paired O switch and contraction	`pairedAuxiliaryContraction`; new combinatorial	Proved
Exact transport and deletion of a singleton	`auxiliarySingletonTransport`; new combinatorial	Proved
Four-mark block normalization	`fourMarkBlockCollapse`; new combinatorial	Proved

Result	Proposed short name and type	Status
Signed relations, shared endpoints, subdivisions, and guard	`signedWordGridRelations`; new combinatorial	Proved
Marked inverse cancellation with a protected grid	`markedInverseCancellation`; new combinatorial	Proved
Common strand by finite Hurwitz moves	`commonStrandHurwitz`; new combinatorial	Proved
Star scissors and explicit rooted forest	`scissorIncidenceForest`; new combinatorial	Proved
Triple preparation, strict seam coordinates, and cyclic exposure	`cyclicSeamCoordinates`; new combinatorial	Proved
Successive factor cleanup and exact auxiliary deletions	`seamFactorCleanup`; new combinatorial	Proved
Simple-factor functional and binary seam reduction	`simpleFactorProjection`; new combinatorial, uses binary seam	Proved
Exact ramification count	`exteriorRamificationCount`; uses the word lengths and invariant identities	Proved
Ordered interchange of common-strand factors	`orderedCommonStrandHurwitz`; new combinatorial	Proved
Statement E	`ExteriorNativeMapStatement`; uses the preceding results and the admitted inputs	Proved

The external source ledger is short. The attached brief [1] is used for exactly the inputs listed in Section 1: assembly and invariant facts; the central and branch event words; the native complex and canonical-image theorem; the supplied commutation and corner-reduction facts; and the binary seam–tensor equivalence. These are hypotheses supplied for the problem. The formulas for incoming and outgoing local domains were checked against Definitions 13.3.7, 13.3.10, 13.4.2, and 13.4.3 of Ozsváth–Stipsicz–Szabó [2]; their chain, grading, colour, and class properties needed here are verified directly in this document. No theorem identifying grids of isotopic links, no Markov or Reidemeister theorem, no surface movie theorem, and no comparison with holomorphic knot Floer homology enters the proof.

References

[1] *Problem brief: the exterior native map in the Lean proof of Lê's conjecture*, supplied Markdown attachment, prepared 2 November 2026. The input assumptions and the target Statement E are those of Sections 2, 3.1, and 5.

[2] P. S. Ozsváth, A. I. Stipsicz, and Z. Szabó, *Grid Homology for Knots and Links*, American Mathematical Society, 2015. Author-hosted preliminary text: https://web.math.princeton.edu/~petero/GridHomologyBook.pdf. Only the explicitly identified local domain definitions are used for a convention check; the needed identities are proved here.